

Supplementary appendix

Locally robust commensurate Bayesian additive regression trees for historical information borrowing

This appendix provides the posterior calculations, ESS derivation, regional theoretical results and supporting leaf calculations, survival formulation, full simulation results and application sensitivities. Main-text equation and theorem references are identified explicitly. The prior ESS reference and the implemented prior-scale selection procedure are described separately.

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A Posterior computation

A.1 Priors, conditioning and numerical settings

Historical and trial-control observations retain their source labels from the main model. Their means are $f(x_{2,i})$ and $f(x_{1,i}) + g(x_{1,i})$, with residual variances σ_2^2 and σ_1^2 , respectively. Fixed centering constants are suppressed below and restored for predictions. The tree priors are independent across trees; conditional on trees and shared hyperparameters, terminal-node parameters follow the component-specific priors in the main model. Posterior dependence is retained by Bayesian backfitting.

For control group $s \in \{1, 2\}$, the residual prior is

$$\sigma_s^2 \sim \text{IG}\left(\frac{v_\sigma}{2}, \frac{v_\sigma \lambda_{\sigma,s}}{2}\right), \quad v_\sigma = 3, \quad (\text{A.1})$$

where the second argument is a scale and $p(u) \propto u^{-a-1} \exp(-b/u)$. If R_y is the range of the relevant centered fitting outcomes, the standard terminal-node rules are $\lambda_f^2 = R_y^2/(16H_f)$ and $\tau_1^2 = R_y^2/(16H_g)$. A positive fallback is used when an empirical range or preliminary residual estimate is degenerate. Each $\lambda_{\sigma,s}$ is a preliminary residual-variance estimate times $\chi_{3,0.10}^2/3$. Gaussian preliminary estimates use linear regression; survival estimates use lognormal AFT regression, with the script's dispersion fallback. These are data-dependent prior scale choices and are held fixed within sampling.

Table S1: lrcBART prior and computation settings.

Quantity	Setting
Historical and discrepancy trees	Simulations: $H_f = 50$, $H_g = 5$; application: $H_f = 10$ or 50 , $H_g = 5$
Treatment trees	$H_{\text{trt}} = 50$ in all studies, including application settings with $H_f = 10$
Depth prior	$(\alpha_f, \beta_f) = (\alpha_{\text{trt}}, \beta_{\text{trt}}) = (0.95, 2)$; $(\alpha_g, \beta_g) = (0.5, 3)$
Terminal-node scale	$\lambda_f^2 = R_y^2/(16H_f)$; $\tau_1^2 = R_y^2/(16H_g)$; $\lambda_{\text{trt}}^2 = R_{y,\text{trt}}^2/(16H_{\text{trt}})$
Residual variance	Inverse-gamma, $v_\sigma = 3$; scale from preliminary residual estimate
Spike variance, two-arm fit	Scaled inverse-chi-square with $v_0 = 3$, restricted below τ_1^2 ; updated
Spike variance, single-arm fit	Fixed at $\min(s_0^2, \tau_1^2/2)$
Mixture probability	Two-arm: Beta(1, 1), updated; single-arm: fixed 1 or 0.9
Tree proposals	Grow, prune, change with probabilities 0.3, 0.3, 0.4
Minimum fitting leaf size	f : 5 controls in total; g : 5 trial controls in two-arm fits, 0 in single-arm fits
Simulation lrcBART sampling	1,000 warm-up; 1,000 retained draws; thinning 1
Application lrcBART sampling	Four chains; 2,000 warm-up and 2,000 retained per chain
Preliminary historical fit	Simulations: 1,000 warm-up and 1,000 retained; application: 2,000 warm-up and 4,000 retained
Prior ESS Monte Carlo	2,000 prior trees, 4,000 scale draws, 20 variance blocks
Scale grid and selector	141 log-spaced values from 10^{-6} to 10; block acceptance fraction 0.95

The sampler overrides the discrepancy minimum leaf size to zero when no trial controls are present. The calibration prior-tree simulation has its own admissibility and maximum-depth rule, discussed in Appendix B.7. Historical and treatment priors use their respective outcome ranges.

A.2 Partial residuals

For the current prognostic tree h^f or discrepancy tree h^g , remove its contribution from the respective sum. Conditional on every other tree and current variance values, form

$$\begin{aligned} r_{2,i}^f &:= Y_{2,i} - f^{(-h^f)}(x_{2,i}), \\ r_{1,i}^f &:= Y_{1,i} - f^{(-h^f)}(x_{1,i}) - g(x_{1,i}), \\ r_{1,i}^g &:= Y_{1,i} - f(x_{1,i}) - g^{(-h^g)}(x_{1,i}). \end{aligned} \tag{A.2}$$

The residuals with subscript 1 use trial controls only. $f^{(-h^f)}$ and $g^{(-h^g)}$ denote the respective sums with the current tree omitted; the tree index is suppressed on the residuals. These residuals remain fixed when comparing current and proposed structures. For survival data, $Y_{2,i}$ and $Y_{1,i}$ in these calculations are the current observed or augmented log event time.

A.3 Marginal leaf likelihoods

Within a specified leaf, write r_i for the relevant partial residuals, suppressing their source labels only in the generic normal integral. For an f leaf these include both $r_{2,i}^f$ and $r_{1,i}^f$, with their respective residual variances; for a g leaf they include only $r_{1,i}^g$.

Lemma A.1 (Normal leaf integral). *Suppose $r_i \mid \theta \sim N(\theta, \sigma_i^2)$ independently in a leaf and $\theta \sim N(0, q^2)$, with $q^2 > 0$. Let $A := \sum_i \sigma_i^{-2}$ and $B := \sum_i r_i \sigma_i^{-2}$. The marginal leaf density is*

$$p(\mathbf{r}_\ell \mid \mathcal{T}_{h^k}^k, \text{rest}) = \left[\prod_{i \in \ell} \phi(r_i; 0, \sigma_i^2) \right] (1 + q^2 A)^{-1/2} \exp \left\{ \frac{q^2 B^2}{2(1 + q^2 A)} \right\}. \tag{A.3}$$

Here $k \in \{f, g\}$, $\mathcal{T}_{h^k}^k$ denotes the current component-specific tree, and θ denotes its leaf parameter. The last argument of ϕ is variance; the result also holds for an empty leaf, using $A = B = 0$ and an empty product of one.

Proof. Factor the observation-level product from the normal likelihood and integrate the remaining prior-weighted kernel:

$$\frac{1}{\sqrt{2\pi q^2}} \int \exp \left\{ -\frac{1}{2} (A + q^{-2}) \theta^2 + B \theta \right\} d\theta. \tag{A.4}$$

Completing the square gives mean $B/(A + q^{-2})$ and precision $A + q^{-2}$. Its normalizing integral is $\{q^2(A + q^{-2})\}^{-1/2} \exp\{B^2/[2(A + q^{-2})]\}$, which is the stated factor. $\square$

For an f leaf, let $n_{s,\ell}, S_{s,\ell}$ be the count and sum of its partial residuals from source s . Write A and B for the leaf precision and precision-weighted residual sum. The marginal likelihood factor is

$$\begin{aligned} A &:= \frac{n_{1,\ell}}{\sigma_1^2} + \frac{n_{2,\ell}}{\sigma_2^2}, & B &:= \frac{S_{1,\ell}}{\sigma_1^2} + \frac{S_{2,\ell}}{\sigma_2^2}, \\ M_f &= (1 + \lambda_f^2 A)^{-1/2} \exp \left\{ \frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)} \right\}. \end{aligned} \tag{A.5}$$

For a g leaf, redefine A and B using only its trial-control count and partial-residual sum. Integrate its parameter under each component, then sum over its indicator:

$$\begin{aligned} A &:= \frac{n_{1,\ell}}{\sigma_1^2}, & B &:= \frac{S_{1,\ell}}{\sigma_1^2}, \\ M_g &= w(1 + \tau_0^2 A)^{-1/2} \exp\left\{\frac{\tau_0^2 B^2}{2(1 + \tau_0^2 A)}\right\} \\ &\quad + (1 - w)(1 + \tau_1^2 A)^{-1/2} \exp\left\{\frac{\tau_1^2 B^2}{2(1 + \tau_1^2 A)}\right\}. \end{aligned} \quad (\text{A.6})$$

Thus both $\theta_{hs\ell}^g$ and $z_{hs\ell}$ are marginalized for a discrepancy-tree proposal. The quantities w, τ_0^2, τ_1^2 are held fixed, not integrated out in this step. For an empty trial-control leaf, $M_g = w + (1 - w) = 1$.

A.4 Metropolis–Hastings tree updates

The marginal tree likelihood is the observation-level density product times the product of its leaf factors. The observation product is identical under both partitions, so it cancels from the Metropolis–Hastings ratio. Suppressing the tree index and component label, a proposal $\mathcal{T} \rightarrow \mathcal{T}^*$ is accepted with probability

$$\min\left\{1, \frac{p(\mathcal{T}^*)}{p(\mathcal{T})} \frac{q(\mathcal{T} | \mathcal{T}^*)}{q(\mathcal{T}^* | \mathcal{T})} \frac{\prod_{\ell \in \mathcal{T}^*} M_\ell}{\prod_{\ell \in \mathcal{T}} M_\ell}\right\}. \quad (\text{A.7})$$

The tree prior includes split/no-split probabilities and admissible variable/cutpoint rules; the proposal ratio accounts for the choice of move, node and rule. For a grow move replacing parent P with children L, R , the likelihood contribution reduces to $M_L M_R / M_P$. The current implementation uses grow, prune and change proposals; an inadmissible proposal is rejected.

A.5 Conditional parameter distributions

After accepting or rejecting a structure, sample node parameters for the retained tree. For an f leaf,

$$\theta_{hf\ell}^f | \text{rest} \sim \text{N}\left(\frac{B}{A + \lambda_f^{-2}}, \frac{1}{A + \lambda_f^{-2}}\right). \quad (\text{A.8})$$

For a nonempty g leaf, its spike probability is

$$P(z_{hs\ell} = 1 | \text{rest}) = \frac{w \phi(\bar{r}; 0, \tau_0^2 + \sigma_1^2/n_{1,\ell})}{w \phi(\bar{r}; 0, \tau_0^2 + \sigma_1^2/n_{1,\ell}) + (1 - w) \phi(\bar{r}; 0, \tau_1^2 + \sigma_1^2/n_{1,\ell})}. \quad (\text{A.9})$$

This compares the density of the partial-residual mean under the two integrated components, including their prior weights. Given component variance τ_k^2 ($k = 0$ for $z = 1$ and $k = 1$ for $z = 0$),

$$\theta_{hs\ell}^g | z_{hs\ell}, \text{rest} \sim \text{N}\left(\frac{B}{A + \tau_k^{-2}}, \frac{1}{A + \tau_k^{-2}}\right). \quad (\text{A.10})$$

An empty leaf instead draws $z \sim \text{Bernoulli}(w)$ and θ from its selected prior component; division by $n_{1,\ell} = 0$ is never required.

Let L_g be the total number of discrepancy terminal nodes, $K_0 := \sum_{h^g, \ell} z_{h^g \ell}$ and $S_0 := \sum_{h^g, \ell} z_{h^g \ell} (\theta_{h^g \ell}^g)^2$. With $w \sim \text{Beta}(a_w, b_w)$, where $a_w = b_w = 1$ by default,

$$w \mid \text{rest} \sim \text{Beta}(a_w + K_0, b_w + L_g - K_0). \quad (\text{A.11})$$

When the spike variance is updated under the hierarchical prior,

$$\tau_0^2 \mid \text{rest} \sim \text{scaled-Inv } \chi^2 \left(v_0 + K_0, \frac{v_0 s_0^2 + S_0}{v_0 + K_0} \right) \text{ restricted to } (0, \tau_1^2). \quad (\text{A.12})$$

For each observed control source, let n_s denote its total sample size and define its residual sum of squares using the full current mean, RSS_s . Then

$$\sigma_s^2 \mid \text{rest} \sim \text{IG} \left(\frac{v_\sigma + n_s}{2}, \frac{v_\sigma \lambda_{\sigma, s} + \text{RSS}_s}{2} \right). \quad (\text{A.13})$$

The treatment sum of trees is fitted separately under the normal or augmented-normal likelihood. It uses $H_{\text{trt}} = 50$, $(\alpha_{\text{trt}}, \beta_{\text{trt}}) = (0.95, 2)$, independent $\theta_{h^{\text{trt}} \ell}^{\text{trt}} \sim \text{N}(0, \lambda_{\text{trt}}^2)$ with $\lambda_{\text{trt}}^2 = R_{y, \text{trt}}^2 / (16H_{\text{trt}})$, and $\sigma_{\text{trt}}^2 \sim \text{IG}(v_\sigma/2, v_\sigma \lambda_{\sigma, \text{trt}}/2)$. The treatment-specific $\lambda_{\sigma, \text{trt}}$ follows the preliminary residual rule in Section A.1.

A.6 Sampling cycle and the single-arm specialization

Algorithm A1: Bayesian backfitting for the control model.

1. Initialize tree structures, terminal-node parameters, scales and, for censored outcomes, latent log event times.
2. For each f tree, form the partial residuals in (A.2), propose a structure using M_f , and sample the retained tree's node parameters.
3. For each g tree, form trial-control partial residuals, propose a structure using M_g , and sample indicators and node parameters for the retained tree.
4. Update τ_0^2 and w when estimated, and update source-specific residual variances.
5. For survival outcomes, redraw censored latent log times above their observed log censoring bounds.
6. Repeat; discard warm-up draws and retain predictions and scalar summaries at the stated thinning interval.

Node draws occur even if a structure proposal is rejected. Logarithmic evaluation and a log-sum-exp calculation for the mixture factor avoid numerical overflow.

When there are no trial controls, every g leaf has $M_g = 1$. Tree structures and leaf parameters for g are then sampled from their specified prior, with no outcome-likelihood update. The sampler permits such prior-only trees by setting their minimum trial-control leaf size to zero. It also sets $\sigma_1^2 = \sigma_2^2$ at each draw. The shared cutpoint specification and any fixed empirical prior scales are treated as conditioned-on quantities. Consequently the single-arm control prediction is $f + g$, with a prior-driven discrepancy.

B Prior effective sample size: comparison, derivation and computation

B.1 Comparison with established prior-ESS definitions

Table S2 places the proposed definition beside the variance calibration of Neuenschwander et al. (2010), the curvature match of Morita et al. (2008), and the expected local-information-ratio (ELIR) definition of Neuenschwander et al. (2020). Throughout the table, D_H denotes the historical dataset, θ_h denotes a historical-study parameter or estimand, with h indexing historical studies when applicable, and θ_* denotes the corresponding current-study parameter or estimand. The two Neuenschwander rows separate the complete-pooling calibration anchor from the partial-pooling MAP prior whose ESS is inferred. For Morita’s method, $Y_{1:m}$ denotes a hypothetical sample of size m from the reference sampling model; write $i_\varepsilon(\theta_*) := -\partial_{\theta_*}^2 \log \pi_\varepsilon(\theta_*)$ for low-information-prior curvature and $j_m(Y_{1:m}; \theta_*) := -\partial_{\theta_*}^2 \log \{\prod_{r=1}^m p(Y_r | \theta_*)\}$ for observed likelihood information. For the proposed method, the target prior is the RWD-informed prior for the current-control mean, and $(\theta_h, \delta_*, \theta_*)$ corresponds to (μ_f, μ_g, μ) in the main text.

Table S2: Comparison of prior effective sample size definitions. Panel A summarizes each target; Panel B gives the corresponding reference or calibration and ESS.

Method/case	Target		
	Target prior	Information measure	Information quantity
Neuenschwander et al. (2010): complete pooling	$\pi_0(\theta_\star) := p(\theta_\star \mid D_H, \tau = 0)$, the complete-pooling MAP prior	Inverse marginal variance, where $V_0 := \text{Var}_{\pi_0}(\theta_\star)$	$\mathcal{I}_0 := 1/V_0$, the complete-pooling prior information
Neuenschwander et al. (2010): partial pooling	$\pi_{\text{target}}(\theta_\star) := p(\theta_\star \mid D_H)$, the partial-pooling MAP prior	Inverse marginal variance, where $V_{\text{target}} := \text{Var}_{\pi_{\text{target}}}(\theta_\star)$	$\mathcal{I}_{\text{target}} := 1/V_{\text{target}}$, the partial-pooling prior information
Morita–Thall–Müller (2008)	Supplied target prior $\pi_{\text{target}}(\theta_\star)$	Negative log-prior curvature at $\bar{\theta}_\star := \text{E}_{\pi_{\text{target}}}(\theta_\star)$	$\mathcal{I}_{\text{target}} := -\partial_{\theta_\star}^2 \log \pi_{\text{target}}(\theta_\star) \Big _{\theta_\star = \bar{\theta}_\star}$
ELIR	Supplied target prior $\pi_{\text{target}}(\theta_\star)$	Pointwise negative log-prior curvature	$\mathcal{I}_{\text{target}}(\theta_\star) := -\partial_{\theta_\star}^2 \log \pi_{\text{target}}(\theta_\star)$
Proposed conditional variance match	RWD-informed target prior $\pi_{\text{target}}(\theta_\star \mid w, \tau_0^2)$ induced for $\theta_\star := \theta_h + \delta_\star$ by the historical posterior and discrepancy prior	Inverse conditional marginal variance, where $V_{\text{target}}(w, \tau_0^2) := \text{Var}_{\pi_{\text{target}}}(\theta_\star \mid w, \tau_0^2)$	$\mathcal{I}_{\text{target}}(w, \tau_0^2) := 1/V_{\text{target}}(w, \tau_0^2)$

Table S2 (continued): Panel B, reference and ESS

Method/case	Reference			ESS	
	Information source	Information measure	Information quantity	Matching or calibration	Result
Neuenschwander: complete pooling	–	–	–	Anchor $\text{ESS}_0 := N_H$, where N_H is the total historical sample size	$\text{ESS}_0 = N_H$
Neuenschwander: partial pooling	–	–	–	$\mathcal{J}_{\text{target}} = (\text{ESS}_{\text{target}}/N_H) \cdot \mathcal{J}_0$	$\text{ESS}_{\text{target}} = N_H V_0 / V_{\text{target}}$
Morita–Thall– Müller	$\pi_{\varepsilon, m}(\theta_* Y_{1:m}) \propto \pi_{\varepsilon}(\theta_*) \prod_{r=1}^m p(Y_r \theta_*)$	Expected negative log-posterior curvature at $\bar{\theta}_*$	$\mathcal{J}_{\text{ref}}(m) := i_{\varepsilon}(\bar{\theta}_*) + E_{Y_{1:m}} \{j_m(Y_{1:m}; \bar{\theta}_*)\}$	$\mathcal{J}_{\text{target}} \approx \mathcal{J}_{\text{ref}}(m)$; choose the closest match	$\text{ESS}_{\text{MTM}} := \arg \min_{m \geq 0} \mathcal{J}_{\text{target}} - \mathcal{J}_{\text{ref}}(m) $
ELIR	One observation $Y \sim p(\cdot \theta_*)$	Expected Fisher information in one observation	$\mathcal{J}_{\text{ref}}(\theta_*) := i_F(\theta_*) = E_{Y \theta_*} \{-\partial_{\theta_*}^2 \log p(Y \theta_*)\}$	$\mathcal{J}_{\text{target}}(\theta_*) = m(\theta_*) \mathcal{J}_{\text{ref}}(\theta_*)$	$\text{ESS}_{\text{ELIR}} := E_{\pi_{\text{target}}} \{m(\theta_*)\}$
Proposed method	One observation $Y \sim p(\cdot \theta_*)$	Expected Fisher information in one observation	$\mathcal{J}_{\text{ref}}(\theta_*) := i_F(\theta_*)$	$\mathcal{J}_{\text{target}}(w, \tau_0^2) = m(w, \tau_0^2; \theta_*) \mathcal{J}_{\text{ref}}(\theta_*)$	$\text{ESS}(w, s_0^2) := E_{\tau_0^2} \{m(w, \tau_0^2; \theta_*)\}$; for the normal reference, $i_F = 1/\sigma_1^2$

Neuenschwander’s complete-pooling prior supplies a known ESS anchor. Morita matches the target-prior curvature to the expected curvature of a posterior constructed from a low-information prior and hypothetical data. ELIR averages local ratios of prior curvature to one-observation Fisher information over the target prior. For the proposed method, this target prior is the RWD-informed prior; the ESS averages conditional inverse-variance matches over the calibration distribution of τ_0^2 .

B.2 Calculation of historical uncertainty

For the main-text historical component $\mu_f = n_1^{-1} \sum_{i=1}^{n_1} f(x_{1,i})$,

$$V_f = \frac{1}{n_1^2} \sum_{i=1}^{n_1} \sum_{i'=1}^{n_1} \text{Cov}\{f(x_{1,i}), f(x_{1,i'}) \mid \mathbf{Y}_2, \mathbf{X}_2, \mathbf{X}_1\}. \quad (\text{B.1})$$

Given preliminary posterior draws $f^{(b)}$, form $\mu_f^{(b)} := n_1^{-1} \sum_i f^{(b)}(x_{1,i})$ and estimate V_f by their sample variance. This calculation integrates tree and leaf uncertainty and retains cross-profile covariance. Averaging the pointwise posterior variances would omit the covariance terms.

B.3 Derivation of discrepancy uncertainty

Let $\mathcal{T}^g := (\mathcal{T}_{h^g}^g)_{h^g=1}^{H_g}$ and define

$$a_{h^g \ell} := \frac{1}{n_1} \sum_{i=1}^{n_1} \mathbf{1}\{x_{1,i} \in \ell \text{ of tree } h^g\}. \quad (\text{B.2})$$

Regrouping the randomized-control-profile average of the discrepancy trees gives

$$\mu_g = \sum_{h^g, \ell} a_{h^g \ell} \theta_{h^g \ell}^g. \quad (\text{B.3})$$

Within each tree, $a_{h^g \ell} \geq 0$ and $\sum_{\ell} a_{h^g \ell} = 1$. Conditional on the trees and indicators, the independent zero-mean leaf priors imply

$$\begin{aligned} \mathbb{E}(\mu_g \mid \mathcal{T}^g, \mathbf{z}, w, \tau_0^2) &= 0, \\ \text{Var}(\mu_g \mid \mathcal{T}^g, \mathbf{z}, w, \tau_0^2) &= \sum_{h^g, \ell} a_{h^g \ell}^2 \{z_{h^g \ell} \tau_0^2 + (1 - z_{h^g \ell}) \tau_1^2\}. \end{aligned} \quad (\text{B.4})$$

The conditional mean remains zero after integrating over the indicators, so the between-mean term in the law of total variance vanishes. Since $\mathbb{E}(z_{h^g \ell} \mid w) = w$,

$$\text{Var}(\mu_g \mid \mathcal{T}^g, w, \tau_0^2) = \{w \tau_0^2 + (1 - w) \tau_1^2\} \sum_{h^g, \ell} a_{h^g \ell}^2.$$

The same argument applies when integrating over the prior trees, giving main-text equation 9. Thus V_g integrates leaf, indicator and tree uncertainty at fixed w, τ_0^2 . The identity is exact under the stated independent zero-mean priors; Monte Carlo enters only in evaluating the tree expectation.

B.4 Proof of the prior ESS curve theorem

Under the all-spike calibration, let

$$C_g := \mathbb{E}_{\mathcal{T}^g} \left[\sum_{h^g, \ell} a_{h^g \ell}^2 \right].$$

Every finite tree has at least one occupied terminal node and $\sum_{\ell} a_{h^g \ell}^2 \leq 1$, so $0 < C_g \leq H_g$. If $U \sim \chi_{v_0}^2$, the scaled inverse-chi-square variable can be coupled over all scale values as $\tau_0^2 = v_0 s_0^2 / U$. Main-text equation 12 becomes

$$\text{ESS}_{\tau_0}(s_0^2) = \mathbb{E}_U \left[\frac{\sigma_1^2}{V_f + C_g v_0 s_0^2 / U} \right].$$

For every $U > 0$, the integrand is continuous, strictly decreasing and strictly convex in $s_0^2 > 0$, and is bounded above by σ_1^2 / V_f . Dominated convergence establishes continuity and the limits

$$\lim_{s_0^2 \downarrow 0} \text{ESS}_{\tau_0}(s_0^2) = \frac{\sigma_1^2}{V_f}, \quad \lim_{s_0^2 \rightarrow \infty} \text{ESS}_{\tau_0}(s_0^2) = 0.$$

Pointwise strict inequalities preserve strict monotonicity and strict convexity after expectation. The intermediate value theorem and strict monotonicity give a unique positive solution for each target in $(0, \sigma_1^2 / V_f)$. This proves main-text Theorem 1 without requiring $\mathbb{E}(\tau_0^2)$ to be finite.

B.5 Monte Carlo evaluation and implemented prior-scale selection

Estimate V_f from the posterior averages of f over the standardization profiles and estimate C_g by simulating discrepancy trees from the calibration prior. For independent $U_b \sim \chi_{v_0}^2$, $b = 1, \dots, B$, evaluate

$$\widehat{\text{ESS}}_{\tau_0}(s_0^2) := \frac{1}{B} \sum_{b=1}^B \frac{\widehat{\sigma}_1^2}{\widehat{V}_f + (v_0 s_0^2 / U_b) \widehat{C}_g}. \quad (\text{B.5})$$

Reusing the U_b values over the scale grid makes the Monte Carlo curve monotone. The preliminary f predictions are posterior draws conditional on the historical dataset.

Algorithm B1: full-variance ESS evaluation and target selection.

1. Specify the current-control standardization profiles, historical outcome likelihood, candidate splitting rules and fixed reference residual variance estimate.
2. Fit the historical model, average each posterior prediction draw over the standardization profiles and estimate V_f .
3. Simulate discrepancy trees from the calibration prior and estimate C_g .
4. Draw $U_b \sim \chi_{v_0}^2$ and calculate (B.5) over candidate scales.
5. For an attainable target, solve the monotone equation or select a grid approximation; record targets outside the attainable range.

The implemented selector divides the posterior draws of μ_f into $J = 20$ consecutive blocks, calculates a variance in each block and rescales the block variances so that their arithmetic mean equals the variance

across all draws. Write the rescaled values as $\widehat{V}_{f,k}$. It substitutes each $\widehat{V}_{f,k}$ into (B.5), yielding $\widehat{\text{ESS}}_k(s_0^2)$, and uses their arithmetic mean as the central curve. With $m_* := \min(m_{\text{requested}}, \widehat{\sigma}_1^2 / \widehat{V}_f)$, the selector chooses the smallest supplied scale satisfying

$$\frac{1}{J} \sum_{k=1}^J \mathbf{1}\{\widehat{\text{ESS}}_k(s_0^2) \leq m_*\} \geq 0.95. \quad (\text{B.6})$$

If no grid value satisfies the rule, it chooses the largest value. If the requested target reaches or exceeds the full-variance ceiling, the current implementation chooses the smallest grid value.

The block-based central curve does not generally have the full-variance ceiling as its zero-scale limit because

$$\lim_{s_0^2 \downarrow 0} \frac{1}{J} \sum_{k=1}^J \widehat{\text{ESS}}_k(s_0^2) = \frac{1}{J} \sum_{k=1}^J \frac{\widehat{\sigma}_1^2}{\widehat{V}_{f,k}} \geq \frac{\widehat{\sigma}_1^2}{\widehat{V}_f},$$

with equality only when the positive block variances are equal. Consecutive blocks of an autocorrelated Markov chain also do not supply a formal 95% posterior or repeated-sampling guarantee. Figure S1 illustrates the curves for one Gaussian two-arm dataset; Table S11 evaluates the full-variance expression in the application.

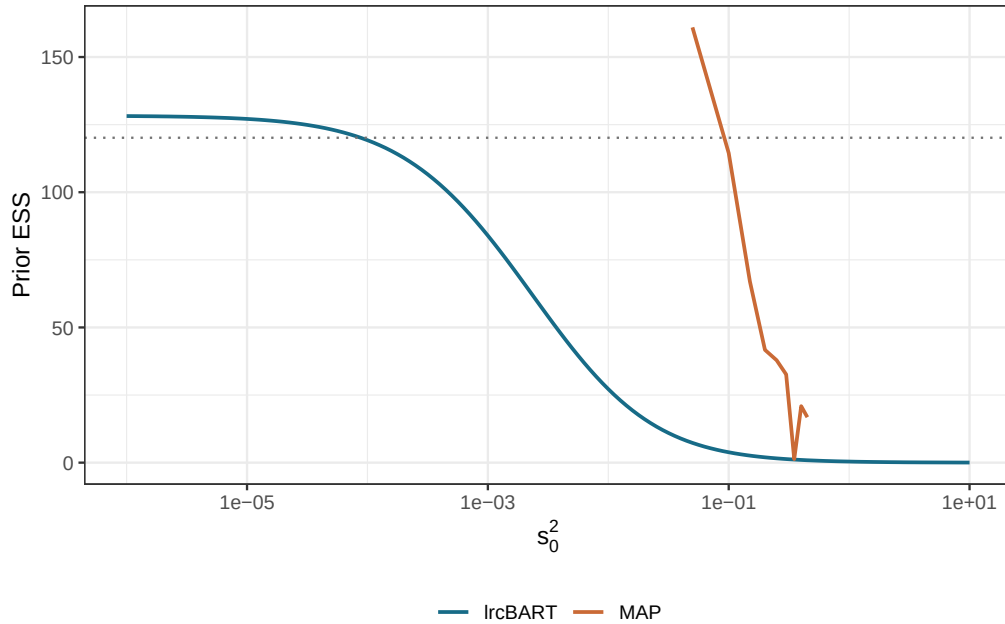


Figure S1: Illustrative prior ESS curves for Gaussian two-arm Sc1, replicate 1. The lrcBART curve is the block-based central ESS summary over the variance-scale grid; the dotted horizontal line is its full-variance ceiling. The MAP overlay uses its corresponding prior ESS calculation. Because MAP and lrcBART use different prior constructions and ESS implementations, the common horizontal axis denotes each method's supplied variance-scale parameter rather than identical prior distributions. The curves use one dataset and are not averaged over replicates.

B.6 Pointwise and overall means

For one standardization profile x , define the scalar current-control target $\mu(x) := f(x) + g(x)$ and

$$V_f(x) := \text{Var}\{f(x) \mid \mathbf{Y}_2, \mathbf{X}_2, x\}.$$

One terminal node in each discrepancy tree has weight one, so

$$V_g(w, \tau_0^2; x) = H_g\{w\tau_0^2 + (1-w)\tau_1^2\}.$$

Under the all-spike reference, the pointwise ESS is therefore

$$\text{ESS}_{\tau_0}(s_0^2; x) = E_{\tau_0^2} \left[\frac{\sigma_1^2}{V_f(x) + H_g\tau_0^2} \right].$$

The matched reference observation has the same target, $Y^{\text{ref}} \mid x \sim N\{\mu(x), \sigma_1^2\}$. Thus the target and reference describe the same conditional mean parameter.

For standardization profiles $x_1, \dots, x_n$, the overall target is $\bar{\mu} = n^{-1} \sum_i \mu(x_i)$ and

$$\text{Var}(\bar{\mu}) = \frac{1}{n^2} \left[\sum_i \text{Var}\{\mu(x_i)\} + \sum_{i \neq i'} \text{Cov}\{\mu(x_i), \mu(x_{i'})\} \right]. \quad (\text{B.7})$$

The same fitted functions inform all profiles, so the covariance terms generally do not vanish. Inverse variance is also nonlinear. Pointwise ESS values therefore do not generally sum or average to the overall ESS; the overall ESS must be calculated from the variance of $\bar{\mu}$.

B.7 Reference interpretation and implementation qualifications

The theoretical reference treats σ_1^2 as fixed. Numerical calibration substitutes a fixed estimate $\hat{\sigma}_1^2$; in two-arm analyses this estimate uses randomized controls. A single-arm analysis specifies its standardization profiles separately and uses the RWD-control residual variance as the reference scale under $\sigma_1^2 = \sigma_2^2$.

For survival outcomes, censoring enters the preliminary historical likelihood and therefore affects V_f . The reference information unit remains one uncensored normal observation on the log-time scale.

The preliminary historical fit, calibration tree simulation and final joint fit use different admissible split rules. The current calibration simulator limits depth to 12, while the final two-arm fit also requires at least five randomized controls per discrepancy leaf. Prior ESS calibration uses an untruncated spike-variance distribution; the two-arm sampler uses its upper-truncated counterpart, and single-arm fits fix the spike variance at $\min(s_0^2, \tau_1^2/2)$. Accordingly, the reported prior ESS corresponds to the stated calibration specification.

C Local agreement and regional borrowing properties

C.1 Regional model and scope

The main-text results address two questions: whether the posterior learns practical agreement between control sources, and how borrowing changes estimation of the current-control mean. They concern a joint Gaussian model on fixed regions. The historical and current-control means are both estimated; neither is treated as a known prediction from another tree update. These results are benchmarks for the borrowing construction, rather than guarantees for the fitted sum-of-trees model.

For main-text Theorem 2, let $R_1, \dots, R_J$ be a fixed finite partition. Within region j , independent observations follow

$$Y_{2,ji} \sim N(f_j^*, \sigma_2^2), \quad Y_{1,ji} \sim N(f_j^* + g_j^*, \sigma_1^2),$$

with known positive residual variances. The fitted model uses independent priors across regions, with $f_j \sim N(0, \lambda_j^2)$ independent of $g_j \sim wN(0, \tau_0^2) + (1-w)N(0, \tau_1^2)$. All prior parameters are fixed, with $0 < \lambda_j^2 < \infty$, $0 < w < 1$, and $0 < \tau_0^2 < \tau_1^2 < \infty$. Each source's sample size tends to infinity in every region. No relative growth-rate restriction is needed for this finite-region result.

Main-text Theorem 3 instead uses the flat prior $p(f) \propto 1$ for the historical mean in one region, retaining the proper discrepancy mixture and fixed known variances. This change removes baseline shrinkage from the exact risk comparison. Its posterior is proper when both source sample sizes are positive, as the normal integral in Section C.3 shows. The flat-baseline risk guarantee must not be transferred to the normal baseline prior without a separate argument.

C.2 Proof of main-text Theorem 2

Fix region j , suppress its index on $v_s := \sigma_s^2/n_{s,j}$, and let $\beta_j = (f_j, g_j)^\top$. The sufficient statistic

$$\hat{\beta}_j = \begin{pmatrix} \bar{Y}_{2,j} \\ \bar{Y}_{1,j} - \bar{Y}_{2,j} \end{pmatrix} \quad (\text{C.1})$$

has normal sampling distribution with mean $\beta_j^* = (f_j^*, g_j^*)^\top$ and covariance

$$\Sigma_j = \begin{pmatrix} v_2 & -v_2 \\ -v_2 & v_1 + v_2 \end{pmatrix}. \quad (\text{C.2})$$

As both sample sizes increase, the strong law gives $\hat{\beta}_j \rightarrow \beta_j^*$ almost surely and $\Sigma_j \rightarrow 0$.

Conditional on discrepancy component $k \in \{0, 1\}$, the prior covariance of β_j is $P_{jk} = \text{diag}(\lambda_j^2, \tau_k^2)$. Normal updating gives

$$\begin{aligned} C_{jk} &= (\Sigma_j^{-1} + P_{jk}^{-1})^{-1}, \\ m_{jk} &= C_{jk} \Sigma_j^{-1} \hat{\beta}_j = \hat{\beta}_j - C_{jk} P_{jk}^{-1} \hat{\beta}_j. \end{aligned} \quad (\text{C.3})$$

Because $0 \prec C_{jk} \preceq \Sigma_j$ in the positive-semidefinite order, $C_{jk} \rightarrow 0$. The matrix P_{jk} is fixed and positive definite, so $m_{jk} \rightarrow \beta_j^*$ almost surely. Each component posterior therefore puts probability tending to one in any neighborhood of β_j^* . The posterior mixture has the same property: its probability outside a neighborhood is

at most the larger of the two component probabilities. This argument does not require the mixture allocation probability to converge to zero or one.

A finite union bound over j establishes the simultaneous posterior convergence in the theorem. The bound

$$|(f_j + g_j) - (f_j^* + g_j^*)| \leq |f_j - f_j^*| + |g_j - g_j^*|$$

proves learning of the current-control means. Finally, if $|g_j^*| \neq c$, put $\delta_j = ||g_j^*| - c| > 0$. A posterior draw for which $|g_j|$ and $|g_j^*|$ are on opposite sides of c must satisfy $|g_j - g_j^*| \geq \delta_j$. Posterior concentration makes this event's probability vanish. There are finitely many regions, so the practical-agreement limits hold almost surely in all of them simultaneously. This conclusion concerns posterior probabilities directly and introduces no decision cutoff for $B_{c,j}$.

C.3 Proof of main-text Theorem 3

Joint posterior and borrowing coefficient. Suppress the region index and write $\mu_1 = f + g$, $v_s = \sigma_s^2/n_s$, $V = v_1 + v_2$, and $D = \bar{Y}_1 - \bar{Y}_2$. Integrating the flat-baseline likelihood over f gives

$$\int_{\mathbb{R}} \phi(\bar{Y}_1; f + g, v_1) \phi(\bar{Y}_2; f, v_2) df = \phi(D; g, V). \quad (\text{C.4})$$

Integrating this expression against the two proper discrepancy components gives the positive finite mixture $w\phi(D; 0, V + \tau_0^2) + (1 - w)\phi(D; 0, V + \tau_1^2)$. Thus the joint posterior is proper, and its spike probability is $\gamma(D)$ in main-text equation (16). Under component k ,

$$g \mid D, k \sim N\left(\frac{\tau_k^2}{V + \tau_k^2} D, \frac{V \tau_k^2}{V + \tau_k^2}\right).$$

Define the precision-weighted mean and coefficient

$$T := \frac{v_2 \bar{Y}_1 + v_1 \bar{Y}_2}{V}, \quad a_* := \frac{v_1}{V}. \quad (\text{C.5})$$

Conditional on g and the observations, the posterior mean of μ_1 is $T + a_* g$. Hence its component-specific posterior mean is $\bar{Y}_1 - a_k D$, where $a_k = v_1/(V + \tau_k^2)$. Averaging over the components proves main-text equation (17).

The posterior spike odds are

$$\frac{\gamma(D)}{1 - \gamma(D)} = C \exp(-\kappa D^2), \quad C = \frac{w}{1 - w} \sqrt{\frac{V + \tau_1^2}{V + \tau_0^2}}, \quad \kappa = \frac{1}{2} \left(\frac{1}{V + \tau_0^2} - \frac{1}{V + \tau_1^2} \right) > 0. \quad (\text{C.6})$$

It follows that $\gamma(D)$ decreases strictly with $|D| > 0$ and tends to zero as $|D| \rightarrow \infty$. Because $a_0 > a_1 > 0$, the coefficient $a(D) = a_1 + (a_0 - a_1)\gamma(D)$ has the stated monotonicity and limit. Also $0 < a(D) < a_* = v_1/V$.

Exact risk and benefit under agreement. Let $d = \mu_1 - f$ denote the fixed true discrepancy in the sampling experiment. Under independent normal sampling, T and D are independent, with

$$\mathbb{E}_d(T) = \mu_1 - a_* d, \quad \text{Var}_d(T) = \frac{v_1 v_2}{V}, \quad D \sim N(d, V). \quad (\text{C.7})$$

The posterior-mean estimator can be written as $\hat{\mu}_1 = T + \{a_* - a(D)\}D$. Taking its expectation or squared error proves the bias and risk expressions in main-text equation (18).

At $d = 0$, $a(D)$ is even and D is symmetrically distributed, so $E_0\{a(D)D\} = 0$. The estimator is therefore unbiased. Since $0 < a(D) < a_*$ and $P(D \neq 0) = 1$,

$$\begin{aligned} R(0) &= \frac{v_1 v_2}{V} + E_0[\{a_* - a(D)\}^2 D^2] \\ &< \frac{v_1 v_2}{V} + a_*^2 E_0(D^2) = \frac{v_1 v_2}{V} + \frac{v_1^2}{V} = v_1. \end{aligned} \quad (\text{C.8})$$

This is a strict repeated-sampling mean squared error improvement over the trial-only mean. It is not a statement about posterior interval width in every dataset. The normal-expectation formula for $R(d)$ is continuous in d , by dominated convergence on compact sets of d . Thus a neighborhood of exact agreement also has $R(d) < v_1$, although its size depends on both sampling variances and the prior parameters.

For numerical evaluation, a second risk identity follows by normal integration by parts:

$$R(d) = v_1 + E_d\{a(D)^2 D^2\} - 2v_1 E_d\{a(D) + Da'(D)\}. \quad (\text{C.9})$$

Indeed, $E(\bar{Y}_1 - \mu_1 \mid D) = a_*(D - d)$ and $E_d\{(D - d)h(D)\} = V E_d\{h'(D)\}$ for $h(D) = a(D)D$. The smooth bounded coefficient and its Gaussian-decaying derivative make the integration valid. Treating $a(D)$ as a fixed coefficient would omit the derivative term and would not give the correct risk.

Severe conflict at fixed sample sizes. Write $b(D) = (a_0 - a_1)\gamma(D)$. Then $a(D) = a_1 + b(D)$ and $0 \leq b(D) \leq (a_0 - a_1)C \exp(-\kappa D^2)$. For each fixed integer $q \geq 0$, completing the square gives

$$E_d\{|D|^q e^{-\kappa D^2}\} = O\left[\left(1 + |d|^q\right) \exp\left\{-\frac{\kappa d^2}{1 + 2\kappa V}\right\}\right].$$

Consequently, the bias contribution $E_d\{b(D)D\}$ tends to zero. Comparing the risk formula with the constant-coefficient estimator $\bar{Y}_1 - a_1 D$ also leaves only polynomial multiples of these vanishing expectations. Therefore

$$\begin{aligned} \text{Bias}_d(\hat{\mu}_1) &= -a_1 d + o(1), \\ R(d) &= a_1^2 d^2 + (1 - a_1)^2 v_1 + a_1^2 v_2 + o(1), \end{aligned} \quad |d| \rightarrow \infty. \quad (\text{C.10})$$

Dividing by d and d^2 , respectively, proves the theorem's limits. A finite normal slab therefore leaves unbounded borrowing bias over arbitrarily large true discrepancies at fixed sample sizes.

Increasing trial information is a different limit. With fixed positive prior scales, fixed true d , and bounded v_2 , the inequality $a(D) \leq v_1/\tau_0^2$ gives

$$|\text{Bias}_d(\hat{\mu}_1)| \leq \frac{v_1}{\tau_0^2} E_d|D| \leq \frac{v_1}{\tau_0^2} (|d| + \sqrt{v_1 + v_2}) \rightarrow 0 \quad \text{as } v_1 \downarrow 0. \quad (\text{C.11})$$

This limit does not permit the spike variance to shrink arbitrarily with the sample size.

C.4 Implications for standardized treatment contrasts

Let $p_j \geq 0$, $\sum_j p_j = 1$, be fixed target-region weights. Write $e_j = \hat{\mu}_{1,j} - \mu_{1,j}$ and $b_j = E(e_j)$. The mean squared error of the standardized control estimate is

$$E\left(\sum_j p_j e_j\right)^2 = \left(\sum_j p_j b_j\right)^2 + \sum_{j,k} p_j p_k \text{Cov}(e_j, e_k). \quad (\text{C.12})$$

Under independent regional data and fixed independent regional priors in the flat-baseline benchmark, the off-diagonal covariance terms vanish. If all regions have exact agreement, the regional estimators are unbiased and their aggregate risk is $\sum_j p_j^2 R_j(0) < \sum_j p_j^2 v_{1,j}$, the risk of the standardized trial-only control mean.

For the treatment contrast, write the error of the same independent treatment estimator as e_{trt} , with bias b_{trt} . The contrast error is $e_{\text{trt}} - \sum_j p_j e_j$, so its mean squared error is

$$\text{Var}(e_{\text{trt}}) + \sum_{j,k} p_j p_k \text{Cov}(e_j, e_k) + \left(b_{\text{trt}} - \sum_j p_j b_j\right)^2.$$

Under exact agreement in all regions, the strict control-risk improvement therefore also improves contrast estimation relative to the same treatment estimator with trial-only controls. Under conflict, the weighted control biases must be retained; if the treatment estimator is unbiased, the contrast bias is $-\sum_j p_j b_j$.

These formulas condition on the regional weights and design. Shared estimated mixture parameters and overlapping trees can induce regional dependence, so covariance terms cannot be discarded for the fitted ensemble. Under the learning model of Theorem 2, posterior consistency of all regional control means also implies consistency of their fixed weighted mean. Adding a consistently learned treatment mean yields consistency of the corresponding contrast. An extension from fixed regions and weights to a learned ensemble standardized over empirical trial covariates requires its own design and posterior-concentration argument.

C.5 Supporting properties of a conditional discrepancy leaf

The following two lemmas retain the original leaf-level calculations as supporting results. Fix a discrepancy-tree partition, every other tree contribution, $n \geq 1$ trial-control partial residuals $\mathbf{r}$ in a leaf, $\sigma_1^2 > 0$, $0 < w < 1$, and $0 < \tau_0^2 < \tau_1^2 < \infty$. Let $\bar{r}$ be their mean, and let $z = 1$ denote the spike component. Normal marginalization gives

$$\frac{P(z = 1 \mid \mathbf{r})}{P(z = 0 \mid \mathbf{r})} = \frac{w}{1-w} \sqrt{\frac{\tau_1^2 + \sigma_1^2/n}{\tau_0^2 + \sigma_1^2/n}} \exp\left[-\frac{\bar{r}^2}{2} \left\{ \frac{1}{\tau_0^2 + \sigma_1^2/n} - \frac{1}{\tau_1^2 + \sigma_1^2/n} \right\}\right]. \quad (\text{C.13})$$

For this subsection, write these odds as $C_\ell e^{-\kappa_\ell \bar{r}^2}$, with $C_\ell, \kappa_\ell > 0$, and put $\gamma_\ell(\bar{r}) = P(z = 1 \mid \mathbf{r})$ and $\omega_k = \tau_k^2 / (\tau_k^2 + \sigma_1^2/n)$.

Lemma C.1 (Bounded additional shrinkage in a fixed leaf). *Under the stated conditional leaf model, the difference between the mixture posterior mean and the slab-only posterior mean is bounded as a function of $\bar{r}$ and tends to zero as $|\bar{r}| \rightarrow \infty$.*

Proof. The component means are $\omega_k \bar{r}$. Hence the difference is $\Delta_{\text{mix}}(\bar{r}) = -\gamma_\ell(\bar{r})(\omega_1 - \omega_0)\bar{r}$. Since $\gamma_\ell(\bar{r}) \leq$

$$C_\ell e^{-\kappa_\ell \bar{r}^2},$$

$$|\Delta_{\text{mix}}(\bar{r})| \leq (\omega_1 - \omega_0) C_\ell |\bar{r}| e^{-\kappa_\ell \bar{r}^2} \leq \frac{(\omega_1 - \omega_0) C_\ell}{\sqrt{2e\kappa_\ell}}.$$

The first bound tends to zero in both tails. Relative to the unshrunk likelihood mean $\bar{r}$, the remaining slab shrinkage is $(\omega_1 - 1)\bar{r}$, which is unbounded at fixed n and finite τ_1^2 . $\square$

Lemma C.2 (Conditional learning of a leaf discrepancy). *At fixed sample size, the posterior spike probability decreases strictly with $|\bar{r}| > 0$. If the fixed leaf receives independent $N(d, \sigma_1^2)$ partial residuals, all other quantities remain fixed, and $n \rightarrow \infty$, then for its parameter θ and $|d| \neq c$, $P(|\theta| < c \mid \mathbf{r}) \rightarrow \mathbf{1}\{|d| < c\}$ almost surely. The spike probability need not converge to zero or one.*

Proof. Monotonicity follows from equation (C.13). Almost surely $\bar{r} \rightarrow d$. Each component posterior has mean $\omega_k \bar{r} \rightarrow d$ and variance $(n/\sigma_1^2 + \tau_k^{-2})^{-1} \rightarrow 0$. Thus each component, and their posterior mixture, concentrates at d . Positive separation from the tolerance boundary proves the probability limit. Nevertheless, the spike probability converges to

$$\frac{w\phi(d; 0, \tau_0^2)}{w\phi(d; 0, \tau_0^2) + (1-w)\phi(d; 0, \tau_1^2)},$$

which is strictly between zero and one for finite d . Learning a discrepancy does not uniquely identify its overlapping prior component. $\square$

C.6 Extension to the full model

The regional theorems fix the partition and all prior parameters; the risk theorem additionally uses a flat baseline prior. In the fitted model, tree structures, other contributions, the shared mixture probability, and the spike variance are estimated and dependent, with data-dependent scale choices. The proper Gaussian baseline prior also contributes shrinkage. None of the regional risk or learning results automatically transfers to that setting. A sum of tree functions can have multiple tree decompositions, so the scientifically meaningful target is the total $g(x)$, together with $f(x)$ and $f(x) + g(x)$.

A full-model learning result would require joint posterior concentration of the historical and current-control functions over the target covariate distribution, with adequate support from both sources. Subject to separation from the tolerance boundary, such concentration could justify convergence of posterior practical-agreement probabilities averaged over that population. It would not by itself imply uniform convergence at every profile, exact recovery of tree boundaries, or an estimation-risk improvement.

Without trial controls, the separate treatment likelihood supplies no information about g . Neither a regional theorem nor an ESS calculation supplies that missing information. The regional-discrepancy figure reports finite-sample empirical behavior; Table 7 reports patient-profile prior information and is not evidence that the discrepancy is consistently learned in the single-arm application. Censored-outcome extensions likewise require additional arguments rather than treating imputed log times as fully observed independent data in these proofs.

D Survival outcomes

D.1 Observed-data likelihood

For group $a \in \{\text{trt}, 1, 2\}$, let $T_{a,i}$ denote event time, $C_{a,i}$ censoring time, $\tilde{T}_{a,i} := \min(T_{a,i}, C_{a,i})$ and $\delta_{a,i} := \mathbf{1}(T_{a,i} \leq C_{a,i})$. Conditional noninformative censoring is assumed given the covariates and source. For a source-specific lognormal model with log-time mean $\eta_{a,i}$ and residual standard deviation σ_a , the outcome likelihood contribution is

$$\left[\frac{1}{\tilde{T}_{a,i}} \phi(\log \tilde{T}_{a,i}; \eta_{a,i}, \sigma_a^2) \right]^{\delta_{a,i}} \left[1 - \Phi\left(\frac{\log \tilde{T}_{a,i} - \eta_{a,i}}{\sigma_a}\right) \right]^{1-\delta_{a,i}}. \quad (\text{D.1})$$

The mean functions are $\eta_{\text{trt}}(x) := f_{\text{trt}}(x)$, $\eta_1(x) := f(x) + g(x)$ and $\eta_2(x) := f(x)$, with $\eta_{a,i} := \eta_a(x_{a,i})$. Each group has its residual standard deviation σ_a . Both events and censored observations contribute to parameter inference.

D.2 Data augmentation

For an event, the latent response is $Y_{a,i} = \log \tilde{T}_{a,i}$. For a censored observation, draw

$$Y_{a,i} \mid \text{rest}, \delta_{a,i} = 0 \sim \text{N}(\eta_{a,i}, \sigma_a^2) \text{ restricted to } (\log \tilde{T}_{a,i}, \infty). \quad (\text{D.2})$$

Conditional on these latent responses, use the Gaussian leaf and variance updates in Appendix A. The trial and historical censoring indicators are both used. Prior ESS retains the uncensored normal reference described in Appendix B.7.

D.3 Standardized survival curves and population medians

At each posterior draw, average the individual survival curves over the trial standardization profiles:

$$\bar{S}_a(t) := \frac{1}{N} \sum_{s \in \{\text{trt}, 1\}} \sum_{i=1}^{n_s} \Phi\left(\frac{\eta_a(x_{s,i}) - \log t}{\sigma_a}\right), \quad a \in \{\text{trt}, 1\}. \quad (\text{D.3})$$

Here $\eta_1(x) = f(x) + g(x)$, $\eta_{\text{trt}}(x) = f_{\text{trt}}(x)$; in a single-arm analysis, $\sigma_1 = \sigma_2$. For horizon $t^* > 0$, define $R_a(t^*) := \int_0^{t^*} \bar{S}_a(t) dt$. The population median M_a solves $\bar{S}_a(M_a) = 1/2$; the median-survival ratio is M_{trt}/M_1 . This median of a mixture generally differs from the mean or median of the individual conditional medians. When the true log-time treatment shift is constant and residual distributions are identical between trial arms, the true ratio is $\exp(\text{effect}) = 1.4$.

The simulation analysis uses root-finding with a lower bound 0.05 and an upper cap 30, replacing roots beyond those limits by the relevant bound. These bounds truncate extreme posterior medians. The primary lrcBART median calculations use each arm's own fitted residual standard deviation, with $\sigma_1 = \sigma_2$ under the single-arm equal-variance assumption.

D.4 Restricted mean survival time

For $T \sim \text{lognormal}(\eta, \sigma^2)$,

$$\begin{aligned} E\{\min(T, t^*)\} &= E\{T\mathbf{1}(T \leq t^*)\} + t^*P(T > t^*) \\ &= \exp(\eta + \sigma^2/2)\Phi\left(\frac{\log t^* - \eta - \sigma^2}{\sigma}\right) + t^*\Phi\left(\frac{\eta - \log t^*}{\sigma}\right). \end{aligned} \quad (\text{D.4})$$

The first term follows by completing the square in the truncated lognormal first moment. Averaging (D.4) over trial standardization profiles gives population RMST at each draw. The application uses this analytic expression with $t^* = 5$ years and each arm's residual standard deviation; ratio and difference intervals are obtained from their posterior draws.

The additional simulation RMST summaries use a 120-point equally spaced grid from 0.001 to the restriction horizon and multiply the average survival probability over that grid by the horizon. The horizons are 3 years for the two-arm study and 5 years for the single-arm study. The true per-replicate RMST ratio is calculated under the data-generating model at the same trial standardization profiles and horizon. Unlike the median ratio under a common log-time shift, this restricted ratio generally varies by dataset.

D.5 Supplementary simulation RMST functionals

For both the two-arm and single-arm simulation summaries, each arm's RMST is calculated using that arm's own fitted residual standard deviation. Thus the control curve at posterior draw d uses $\sigma_1^{(d)}$, while the treatment curve uses $\sigma_{\text{trt}}^{(d)}$; no residual-scale substitution is applied.

Simulation ratios replace each control RMST denominator by its maximum with 0.05, while arm-specific summaries use the unfloored values. Tables S5 and S7 report these denominator-floored ratios. Their errors are evaluated against the data-generating RMST ratio at the same profiles and horizon. These supplementary operating characteristics therefore include the effect of the denominator floor.

E Simulation design and complete results

E.1 Covariates, selection and scenario definitions

Each setting has ten observed covariates. Here X_j denotes scalar predictor j , whereas the bold $\mathbf{X}_a$ in the model denotes the covariate collection for group a . For a profile x , x_j denotes coordinate j . Before source selection, the ordinary covariates are independent normals with unit variance and mean vector

$$(1, -1, 0.5, -1, 2, -2, 2, -3, 1, 0). \quad (\text{E.1})$$

With probability 0.15, X_1 is instead drawn from an equally weighted mixture of $N(-3, 0.5^2)$ and $N(3, 0.5^2)$. Separate trial and historical candidate pools are generated. In Sc1, Sc4 and Sc5, source eligibility is independent of covariates. In Sc2, the trial-eligibility probability is

$$\text{logit}^{-1} \left\{ \frac{b_D(X_5 + X_6) - q}{t_D} \right\}, \quad (\text{E.2})$$

where q is the empirical quantile of the pool's selection score at the intended trial fraction. Two-arm studies use $b_D = -1.20, t_D = 0.5$; single-arm studies use $b_D = -0.55, t_D = 1$. Thus the single-arm and two-arm selection strengths differ. Exact numbers of eligible trial and historical observations are then sampled from their respective pools. In two-arm trials, treatment is assigned independently with the probability in main Table 1.

Sc3 is included only in the two-arm studies. Latent U_5, U_6 replace X_5, X_6 in both the outcome function and selection score. They are generated from the corresponding normal distributions. The observed proxies are $X_j = \rho U_j^{\text{std}} + \sqrt{1 - \rho^2} e_j$, where U_j^{std} is standardized within its candidate pool and $e_j \sim N(0, 1)$ independently. We use $\rho = -0.5, 0, 0.5$. Conditioning on these proxies generally does not remove source differences induced by latent-variable selection.

Sc4 adds $\delta \in \{1, 2\}$ to every historical Gaussian outcome or latent log event time. Sc5 adds δ only outside a compatible region: $X_5 > 2$ with $\delta \in \{0.5, 1, 2\}$, or $X_7 \leq 2$ with $\delta \in \{1, 2\}$. The true trial-minus-historical discrepancy is zero inside that region and $-\delta$ outside.

Table S3: Design-specific data-generating constants.

Design	b_0 : Sc1	b_0 : Sc2/3	$\sigma_1 = \sigma_{\text{trt}}$	σ_2	Censoring
Gaussian, two-arm	0.42	-0.08	1.5	1.8	None
Survival, two-arm	-0.20	-0.70	1.4	1.8	$C_d \sim \text{Exp}(0.02)$; $C_a \sim U(2, 3)$
Gaussian, single-arm	0.40	-0.10	1.5	1.5	None
Survival, single-arm	0.20	0.10	1.25	1.25	$C_d \sim \text{Exp}(0.09)$; $C_a \sim U(5, 5.5)$

Sc4 and Sc5 use the Sc1 intercept before adding the historical shift. Sc3 is not run in the single-arm studies. The survival constants apply on the log-time scale, with independent random dropout C_d and administrative censoring C_a . Censoring rates are generated by the outcome and censoring distributions; they are not fixed percentages.

E.2 Outcome functions and censoring

The common nonlinear mean function, before a treatment or historical shift, is

$$\begin{aligned}
 F(x) := & b_0 + \beta_1 x_1^2 + \beta_2 \exp\{-(x_2 + 1)^2/2\} + \beta_3 |x_3 - 0.5| + \beta_4 |x_4 + 1| \\
 & + \beta_5 \tanh(x_5 - 2) + \beta_6 \tanh(x_6 + 2) \\
 & + 2 \left\{ \frac{(x_5 - 2)^2}{1 + ((x_5 - 2)/1.5)^2} + \frac{(x_6 + 2)^2}{1 + ((x_6 + 2)/1.5)^2} \right\} + \sum_{j=7}^{10} \beta_j x_j,
 \end{aligned} \tag{E.3}$$

with $\beta = (-0.50, -0.75, -0.50, -0.50, -1.35, -0.80, -0.01, -0.01, -0.01, -0.01)$. In Sc3, substitute the latent variables in the fifth and sixth positions. For Gaussian outcomes add independent normal noise and a treatment effect of 1 for treated trial participants. For survival outcomes exponentiate the noisy log-time response, with treatment effect $\log(1.4)$. The effect is constant in the reported settings; no additional treatment-by-covariate interaction is generated.

Survival observations are $\min(T, C_d, C_a)$ with event indicator $\mathbf{1}(T \leq C_d, T \leq C_a)$. Two-arm administrative follow-up is generated as 3 minus a uniform entry time on $(0, 1)$; single-arm follow-up is 5.5 minus uniform

entry on $(0, 0.5)$. Historical shifts are applied to latent event times before evaluating censoring. The historical candidate pool is regenerated independently for each replicate.

E.3 Truth and target population

For each dataset, true conditional means and control/treatment survival distributions are evaluated at that dataset’s trial profiles. The Gaussian target is their average treatment-minus-control difference, which equals 1 under the constant-effect design. The survival primary target is the ratio of medians of the two trial-standardized population curves, equal to 1.4 under the generated common log-time shift. The RMST target uses the corresponding restricted integrals and varies with the profiles and restriction horizon. Squared errors are computed against the relevant dataset-specific truth before aggregation.

E.4 Comparator specifications and configuration coverage

Table S4: Methods and reporting configurations.

Method	Specification and included settings
LM / AFT	Parametric outcome regression. Two-arm versions fit trial controls only (-NP), pool controls (-CP), or add a source indicator (-PP). Single-arm -CP gives historical outcomes likelihood weight 1. Treated outcomes are fitted separately.
BART	Corresponding -NP, -CP, or -PP control fits; separate treated surface. Single-arm BART-CP uses historical controls without a discrepancy sum of trees.
Hierarchical LM / AFT	Source-specific intercepts and coefficients have common hypermeans and inverse-gamma hierarchical variances, with discrepancy-variance prior scale $s_\tau^2 = 0.25$ in two-arm studies and $s_\tau^2 = 0.05$ in single-arm studies.
PSCL	Gaussian two-arm only. Propensity-score estimation on the ten observed covariates, distance-based borrowing allocation with total borrowing parameter 100, and composite-likelihood outcome analysis.
MAP	Gaussian two-arm only. Normal study-level mean and standard-error summary of historical controls; MAP construction and normal-mixture approximation; requested ESS 100, 75 or 50. No covariate adjustment in this comparator.
lrcBART, two-arm	$H_f = 50$, $H_g = 5$, estimated w ; ESS targets 100 (primary), 75, 50, f90, f50, f25.
lrcBART, single-arm	$H_f = 50$, $H_g = 5$; fixed $w = 1$ with targets 100 (primary), f90 or s0min; fixed $w = 0.9$ with targets 100 or f90.

f90, f50 and f25 denote fractions of the dataset-specific calibration ceiling. s0min sets $s_0^2 = 10^{-6}$ without target selection. The single-arm target 100 is an absolute count at both survival trial sizes, including $n_{\text{trt}} = 30$.

The hierarchical comparator uses source-specific regression coefficients with a common estimated hypermean. Intercept and predictor-specific discrepancy-variance components have $\text{IG}(3/2, 3s_\tau^2/2)$ priors, where s_τ^2 is the scale listed in Table S3; it is not a probability or historical sample-size fraction. Hypermeans have diffuse normal priors. This comparator pools regression parameters through their prior rather than specifying a nonlinear discrepancy surface.

The MAP comparator uses the historical sample mean and its standard error in a study-level normal model, a normal location prior with standard deviation 10, and an inverse-gamma prior $\text{IG}(3/2, 3a/2)$ for its heterogeneity parameter, with a selected by the MAP prior ESS procedure. Its predictive prior is approximated by a normal mixture and updated using the trial-control sample mean and standard error. This study-level

comparator does not use a BART outcome model. Its prior ESS procedure is distinct from the lrcBART definition. PSCL follows the propensity-score-integrated construction described by Chen et al. (2020).

There are 17 method/configuration combinations per Gaussian two-arm setting and 13 per survival two-arm setting. Each reported single-arm setting has eight combinations. All reported configurations have 100 completed replicates.

E.5 Performance measures and Monte Carlo uncertainty

Let ψ denote the estimand being evaluated: the Gaussian mean contrast, survival median ratio or RMST ratio. For point estimates $\hat{\psi}_r$ and dataset-specific truths ψ_r , write $e_r := \hat{\psi}_r - \psi_r$ over $R = 100$ replicates. We calculate

$$\widehat{\text{Bias}} := R^{-1} \sum_r e_r, \quad \widehat{\text{RMSE}} := \sqrt{R^{-1} \sum_r e_r^2}. \quad (\text{E.4})$$

Gaussian point estimates use posterior means and survival ratios use posterior medians, with the corresponding point estimates used for likelihood-based methods. Coverage is the proportion of reported 95% intervals containing truth. The uncertainty column reports the mean posterior SD for Bayesian methods and the mean standard error for PSCL; interval width is averaged within each setting. These summaries differ from the empirical SD of point estimates across datasets.

For Bayesian benefit decisions, empirical power is the proportion satisfying posterior probability above 0.95 for an effect greater than 0.5 (Gaussian) or a ratio greater than 1 (survival). The PSCL decision instead requires its lower confidence limit to exceed the Gaussian threshold. These are different decision rules. Type I error was not evaluated, and the rules were not calibrated to a common demonstrated Type I error rate.

Monte Carlo standard errors are

$$\widehat{\text{SE}}(\widehat{\text{Bias}}) = \frac{s_e}{\sqrt{R}}, \quad \widehat{\text{SE}}(\widehat{\text{RMSE}}) \simeq \frac{s_{e^2}}{2\sqrt{R}\widehat{\text{RMSE}}}, \quad \widehat{\text{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/R}. \quad (\text{E.5})$$

Here s_e and s_{e^2} are the sample standard deviations of the replicate errors and squared errors, and $\hat{p}$ is the corresponding empirical coverage or power proportion. The RMSE formula is a delta-method approximation. A plug-in proportion standard error of zero at an observed proportion of zero or one does not establish absence of uncertainty; exact binomial intervals would still have nonzero width. These errors describe each row, not a paired between-method contrast. Comparisons should account for the common datasets if formal uncertainty in performance differences is required.

E.6 Gaussian two-arm results

All 12 settings and 17 configurations are reported. Main Table 2 selects Sc1, Sc2, Sc3 with $\rho = 0$, Sc4 with $\delta = 1$, and Sc5 outside $X_5 > 2$ with $\delta = 1$.

Table S5: Complete Gaussian simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: Sc1</i>						
lrcBART: 100	0.003 (0.018)	0.174 (0.012)	0.173	0.676	0.950 (0.022)	0.890 (0.031)
lrcBART: 75	0.008 (0.017)	0.173 (0.011)	0.175	0.684	0.960 (0.020)	0.900 (0.030)
lrcBART: 50	0.001 (0.018)	0.175 (0.011)	0.177	0.694	0.950 (0.022)	0.890 (0.031)
lrcBART: f90	0.003 (0.017)	0.172 (0.011)	0.171	0.669	0.950 (0.022)	0.880 (0.032)
lrcBART: f50	0.004 (0.017)	0.174 (0.011)	0.174	0.681	0.940 (0.024)	0.900 (0.030)
lrcBART: f25	0.002 (0.018)	0.183 (0.012)	0.179	0.700	0.930 (0.026)	0.860 (0.035)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	0.009 (0.015)	0.153 (0.011)	0.146	0.570	0.950 (0.022)	0.980 (0.014)
BART-PP	-0.001 (0.020)	0.197 (0.013)	0.211	0.826	0.950 (0.022)	0.800 (0.040)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	0.015 (0.026)	0.262 (0.019)	0.241	0.946	0.940 (0.024)	0.710 (0.045)
LM-PP	-0.005 (0.035)	0.349 (0.022)	0.341	1.334	0.950 (0.022)	0.390 (0.049)
hier. LM	-0.002 (0.033)	0.329 (0.022)	0.321	1.258	0.940 (0.024)	0.450 (0.050)
PSCL	0.019 (0.029)	0.290 (0.019)	0.267	1.048	0.920 (0.027)	0.480 (0.050)
MAP (100)	0.023 (0.030)	0.302 (0.023)	0.302	1.183	0.940 (0.024)	0.530 (0.050)
MAP (75)	0.023 (0.031)	0.305 (0.022)	0.305	1.195	0.950 (0.022)	0.520 (0.050)
MAP (50)	0.019 (0.032)	0.315 (0.023)	0.315	1.234	0.950 (0.022)	0.490 (0.050)
<i>Mean difference: Sc2</i>						
lrcBART: 100	0.003 (0.019)	0.186 (0.014)	0.180	0.702	0.940 (0.024)	0.860 (0.035)
lrcBART: 75	0.007 (0.019)	0.187 (0.014)	0.180	0.706	0.940 (0.024)	0.890 (0.031)
lrcBART: 50	0.000 (0.018)	0.184 (0.014)	0.181	0.708	0.950 (0.022)	0.870 (0.034)
lrcBART: f90	0.005 (0.019)	0.188 (0.013)	0.182	0.708	0.940 (0.024)	0.860 (0.035)
lrcBART: f50	-0.000 (0.019)	0.192 (0.014)	0.185	0.725	0.950 (0.022)	0.830 (0.038)
lrcBART: f25	-0.005 (0.020)	0.198 (0.014)	0.187	0.731	0.940 (0.024)	0.820 (0.038)
BART-NP	-0.032 (0.021)	0.211 (0.016)	0.203	0.795	0.940 (0.024)	0.750 (0.043)
BART-CP	0.048 (0.018)	0.188 (0.015)	0.176	0.688	0.930 (0.026)	0.950 (0.022)
BART-PP	-0.004 (0.020)	0.200 (0.014)	0.211	0.826	0.970 (0.017)	0.740 (0.044)
LM-NP	-0.062 (0.035)	0.358 (0.025)	0.317	1.246	0.900 (0.030)	0.440 (0.050)
LM-CP	0.343 (0.031)	0.463 (0.028)	0.283	1.108	0.730 (0.044)	0.920 (0.027)
LM-PP	0.007 (0.036)	0.355 (0.027)	0.334	1.304	0.950 (0.022)	0.450 (0.050)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
hier. LM	0.052 (0.036)	0.363 (0.026)	0.320	1.253	0.930 (0.026)	0.510 (0.050)
PSCL	0.007 (0.032)	0.317 (0.022)	0.292	1.145	0.910 (0.029)	0.390 (0.049)
MAP (100)	0.186 (0.042)	0.458 (0.030)	0.384	1.501	0.870 (0.034)	0.550 (0.050)
MAP (75)	0.168 (0.041)	0.444 (0.030)	0.380	1.482	0.860 (0.035)	0.530 (0.050)
MAP (50)	0.140 (0.041)	0.427 (0.029)	0.377	1.471	0.890 (0.031)	0.500 (0.050)
<i>Mean difference: Sc3, $\rho = -0.5$</i>						
lrcBART: 100	0.049 (0.027)	0.269 (0.018)	0.276	1.079	0.960 (0.020)	0.590 (0.049)
lrcBART: 75	0.054 (0.027)	0.273 (0.019)	0.274	1.070	0.950 (0.022)	0.630 (0.048)
lrcBART: 50	0.054 (0.027)	0.274 (0.018)	0.275	1.074	0.950 (0.022)	0.620 (0.049)
lrcBART: f90	0.046 (0.026)	0.267 (0.018)	0.274	1.073	0.960 (0.020)	0.620 (0.049)
lrcBART: f50	0.054 (0.027)	0.273 (0.019)	0.276	1.080	0.940 (0.024)	0.620 (0.049)
lrcBART: f25	0.057 (0.027)	0.271 (0.018)	0.275	1.066	0.960 (0.020)	0.650 (0.048)
BART-NP	0.007 (0.028)	0.276 (0.018)	0.284	1.116	0.950 (0.022)	0.530 (0.050)
BART-CP	0.844 (0.022)	0.872 (0.021)	0.205	0.800	0.010 (0.010)	1.000 (0.000)
BART-PP	0.043 (0.026)	0.260 (0.017)	0.274	1.073	0.980 (0.014)	0.650 (0.048)
LM-NP	-0.005 (0.035)	0.348 (0.023)	0.355	1.390	0.930 (0.026)	0.420 (0.049)
LM-CP	0.870 (0.029)	0.916 (0.026)	0.262	1.025	0.100 (0.030)	0.990 (0.010)
LM-PP	0.089 (0.034)	0.350 (0.022)	0.347	1.361	0.960 (0.020)	0.560 (0.050)
hier. LM	0.136 (0.035)	0.378 (0.023)	0.352	1.374	0.940 (0.024)	0.620 (0.049)
PSCL	0.687 (0.030)	0.747 (0.027)	0.280	1.097	0.320 (0.047)	0.990 (0.010)
MAP (100)	0.254 (0.040)	0.471 (0.029)	0.379	1.480	0.910 (0.029)	0.650 (0.048)
MAP (75)	0.236 (0.040)	0.459 (0.029)	0.375	1.462	0.900 (0.030)	0.640 (0.048)
MAP (50)	0.214 (0.039)	0.443 (0.028)	0.372	1.454	0.910 (0.029)	0.610 (0.049)
<i>Mean difference: Sc3, $\rho = 0$</i>						
lrcBART: 100	0.009 (0.032)	0.315 (0.025)	0.279	1.090	0.930 (0.026)	0.560 (0.050)
lrcBART: 75	0.010 (0.032)	0.315 (0.024)	0.280	1.096	0.940 (0.024)	0.550 (0.050)
lrcBART: 50	0.015 (0.032)	0.315 (0.025)	0.281	1.099	0.950 (0.022)	0.570 (0.050)
lrcBART: f90	0.007 (0.032)	0.317 (0.025)	0.279	1.087	0.950 (0.022)	0.550 (0.050)
lrcBART: f50	0.009 (0.032)	0.314 (0.025)	0.281	1.098	0.930 (0.026)	0.550 (0.050)
lrcBART: f25	0.015 (0.032)	0.316 (0.025)	0.280	1.098	0.940 (0.024)	0.530 (0.050)
BART-NP	-0.046 (0.033)	0.328 (0.025)	0.287	1.125	0.950 (0.022)	0.520 (0.050)
BART-CP	0.957 (0.023)	0.985 (0.024)	0.202	0.789	0.010 (0.010)	1.000 (0.000)
BART-PP	0.004 (0.031)	0.309 (0.024)	0.276	1.080	0.940 (0.024)	0.550 (0.050)
LM-NP	-0.057 (0.036)	0.367 (0.030)	0.355	1.387	0.930 (0.026)	0.340 (0.047)
LM-CP	0.989 (0.028)	1.027 (0.028)	0.254	0.993	0.050 (0.022)	1.000 (0.000)
LM-PP	0.047 (0.035)	0.350 (0.025)	0.350	1.366	0.940 (0.024)	0.500 (0.050)
hier. LM	0.085 (0.036)	0.373 (0.027)	0.358	1.399	0.930 (0.026)	0.510 (0.050)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
PSCL	0.667 (0.028)	0.721 (0.026)	0.273	1.068	0.290 (0.045)	0.980 (0.014)
MAP (100)	0.202 (0.039)	0.434 (0.029)	0.376	1.464	0.910 (0.029)	0.610 (0.049)
MAP (75)	0.182 (0.038)	0.419 (0.028)	0.373	1.459	0.940 (0.024)	0.590 (0.049)
MAP (50)	0.162 (0.037)	0.401 (0.027)	0.370	1.446	0.940 (0.024)	0.590 (0.049)
<i>Mean difference: Sc3, $\rho = 0.5$</i>						
lrcBART: 100	-0.031 (0.027)	0.275 (0.019)	0.274	1.074	0.920 (0.027)	0.560 (0.050)
lrcBART: 75	-0.027 (0.027)	0.272 (0.020)	0.273	1.066	0.920 (0.027)	0.570 (0.050)
lrcBART: 50	-0.026 (0.027)	0.272 (0.020)	0.272	1.064	0.920 (0.027)	0.580 (0.049)
lrcBART: f90	-0.029 (0.027)	0.271 (0.019)	0.271	1.058	0.920 (0.027)	0.580 (0.049)
lrcBART: f50	-0.027 (0.028)	0.275 (0.020)	0.272	1.068	0.920 (0.027)	0.530 (0.050)
lrcBART: f25	-0.024 (0.027)	0.271 (0.019)	0.272	1.059	0.920 (0.027)	0.570 (0.050)
BART-NP	-0.079 (0.027)	0.280 (0.019)	0.281	1.101	0.920 (0.027)	0.470 (0.050)
BART-CP	0.772 (0.021)	0.799 (0.020)	0.205	0.803	0.050 (0.022)	1.000 (0.000)
BART-PP	-0.040 (0.027)	0.269 (0.020)	0.271	1.059	0.920 (0.027)	0.550 (0.050)
LM-NP	-0.098 (0.032)	0.331 (0.024)	0.350	1.370	0.950 (0.022)	0.310 (0.046)
LM-CP	0.806 (0.026)	0.845 (0.024)	0.265	1.037	0.160 (0.037)	1.000 (0.000)
LM-PP	-0.019 (0.033)	0.326 (0.022)	0.348	1.364	0.970 (0.017)	0.460 (0.050)
hier. LM	0.036 (0.033)	0.334 (0.020)	0.348	1.360	0.970 (0.017)	0.530 (0.050)
PSCL	0.578 (0.028)	0.641 (0.025)	0.281	1.100	0.450 (0.050)	0.970 (0.017)
MAP (100)	0.125 (0.041)	0.429 (0.025)	0.374	1.455	0.930 (0.026)	0.570 (0.050)
MAP (75)	0.106 (0.040)	0.410 (0.023)	0.371	1.445	0.940 (0.024)	0.570 (0.050)
MAP (50)	0.085 (0.039)	0.399 (0.023)	0.368	1.439	0.950 (0.022)	0.550 (0.050)
<i>Mean difference: Sc4, $\delta = 1$</i>						
lrcBART: 100	-0.007 (0.020)	0.201 (0.014)	0.192	0.751	0.940 (0.024)	0.810 (0.039)
lrcBART: 75	-0.012 (0.020)	0.197 (0.013)	0.193	0.754	0.940 (0.024)	0.790 (0.041)
lrcBART: 50	-0.014 (0.021)	0.208 (0.015)	0.193	0.751	0.930 (0.026)	0.780 (0.041)
lrcBART: f90	-0.007 (0.020)	0.198 (0.014)	0.193	0.755	0.930 (0.026)	0.820 (0.038)
lrcBART: f50	-0.013 (0.020)	0.201 (0.013)	0.192	0.749	0.930 (0.026)	0.810 (0.039)
lrcBART: f25	-0.015 (0.020)	0.201 (0.013)	0.193	0.753	0.940 (0.024)	0.780 (0.041)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.689 (0.015)	0.706 (0.015)	0.147	0.576	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.012 (0.020)	0.197 (0.013)	0.212	0.832	0.970 (0.017)	0.770 (0.042)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.726 (0.026)	0.772 (0.026)	0.242	0.944	0.180 (0.038)	0.010 (0.010)
LM-PP	-0.054 (0.035)	0.352 (0.020)	0.341	1.334	0.960 (0.020)	0.350 (0.048)
hier. LM	-0.104 (0.034)	0.358 (0.021)	0.330	1.288	0.930 (0.026)	0.290 (0.045)
PSCL	-0.483 (0.029)	0.563 (0.026)	0.268	1.052	0.530 (0.050)	0.060 (0.024)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
MAP (100)	-0.230 (0.041)	0.467 (0.025)	0.360	1.406	0.860 (0.035)	0.170 (0.038)
MAP (75)	-0.204 (0.040)	0.451 (0.026)	0.360	1.404	0.870 (0.034)	0.180 (0.038)
MAP (50)	-0.175 (0.040)	0.433 (0.025)	0.358	1.397	0.910 (0.029)	0.220 (0.041)
<i>Mean difference: Sc4, $\delta = 2$</i>						
lrcBART: 100	0.000 (0.020)	0.196 (0.012)	0.190	0.739	0.960 (0.020)	0.840 (0.037)
lrcBART: 75	0.001 (0.019)	0.194 (0.013)	0.190	0.740	0.930 (0.026)	0.840 (0.037)
lrcBART: 50	-0.005 (0.020)	0.196 (0.013)	0.191	0.745	0.950 (0.022)	0.830 (0.038)
lrcBART: f90	-0.005 (0.020)	0.197 (0.013)	0.190	0.742	0.950 (0.022)	0.820 (0.038)
lrcBART: f50	-0.005 (0.020)	0.196 (0.012)	0.191	0.743	0.960 (0.020)	0.830 (0.038)
lrcBART: f25	-0.009 (0.020)	0.197 (0.013)	0.191	0.745	0.950 (0.022)	0.820 (0.038)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-1.393 (0.016)	1.402 (0.016)	0.153	0.597	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.015 (0.019)	0.194 (0.013)	0.213	0.833	0.980 (0.014)	0.780 (0.041)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-1.466 (0.026)	1.490 (0.026)	0.245	0.957	0.000 (0.000)	0.000 (0.000)
LM-PP	-0.097 (0.035)	0.358 (0.020)	0.342	1.333	0.940 (0.024)	0.340 (0.047)
hier. LM	-0.119 (0.036)	0.377 (0.022)	0.338	1.325	0.920 (0.027)	0.300 (0.046)
PSCL	-0.985 (0.029)	1.027 (0.028)	0.272	1.067	0.080 (0.027)	0.000 (0.000)
MAP (100)	-0.124 (0.040)	0.414 (0.025)	0.366	1.431	0.940 (0.024)	0.300 (0.046)
MAP (75)	-0.112 (0.040)	0.414 (0.025)	0.365	1.427	0.920 (0.027)	0.300 (0.046)
MAP (50)	-0.107 (0.039)	0.405 (0.025)	0.362	1.419	0.940 (0.024)	0.320 (0.047)
<i>Mean difference: Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	-0.048 (0.019)	0.191 (0.011)	0.181	0.705	0.940 (0.024)	0.850 (0.036)
lrcBART: 75	-0.046 (0.018)	0.186 (0.011)	0.181	0.710	0.950 (0.022)	0.810 (0.039)
lrcBART: 50	-0.037 (0.018)	0.187 (0.011)	0.183	0.714	0.960 (0.020)	0.840 (0.037)
lrcBART: f90	-0.050 (0.019)	0.193 (0.011)	0.180	0.705	0.950 (0.022)	0.810 (0.039)
lrcBART: f50	-0.041 (0.018)	0.186 (0.011)	0.182	0.710	0.980 (0.014)	0.800 (0.040)
lrcBART: f25	-0.033 (0.019)	0.188 (0.011)	0.184	0.719	0.980 (0.014)	0.810 (0.039)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.165 (0.015)	0.223 (0.012)	0.146	0.570	0.800 (0.040)	0.720 (0.045)
BART-PP	-0.011 (0.019)	0.194 (0.012)	0.211	0.825	0.960 (0.020)	0.750 (0.043)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.169 (0.026)	0.311 (0.022)	0.241	0.947	0.870 (0.034)	0.410 (0.049)
LM-PP	-0.017 (0.035)	0.346 (0.021)	0.344	1.347	0.950 (0.022)	0.360 (0.048)
hier. LM	-0.032 (0.033)	0.332 (0.021)	0.322	1.260	0.930 (0.026)	0.370 (0.048)
PSCL	-0.105 (0.029)	0.308 (0.020)	0.268	1.051	0.920 (0.027)	0.340 (0.047)
MAP (100)	-0.079 (0.031)	0.323 (0.023)	0.306	1.200	0.950 (0.022)	0.360 (0.048)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
MAP (75)	-0.071 (0.032)	0.328 (0.024)	0.310	1.215	0.930 (0.026)	0.370 (0.048)
MAP (50)	-0.060 (0.033)	0.331 (0.024)	0.317	1.245	0.930 (0.026)	0.360 (0.048)
<i>Mean difference: Sc5, X₅, $\delta = 1$</i>						
lrcBART: 100	-0.050 (0.021)	0.214 (0.012)	0.196	0.766	0.930 (0.026)	0.690 (0.046)
lrcBART: 75	-0.051 (0.021)	0.211 (0.012)	0.195	0.759	0.920 (0.027)	0.720 (0.045)
lrcBART: 50	-0.047 (0.020)	0.207 (0.012)	0.193	0.751	0.950 (0.022)	0.750 (0.043)
lrcBART: f90	-0.055 (0.021)	0.218 (0.013)	0.196	0.764	0.940 (0.024)	0.730 (0.044)
lrcBART: f50	-0.046 (0.021)	0.210 (0.013)	0.194	0.756	0.950 (0.022)	0.720 (0.045)
lrcBART: f25	-0.044 (0.020)	0.204 (0.013)	0.192	0.751	0.950 (0.022)	0.730 (0.044)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.334 (0.015)	0.366 (0.014)	0.147	0.573	0.370 (0.048)	0.350 (0.048)
BART-PP	-0.014 (0.020)	0.195 (0.012)	0.213	0.835	0.970 (0.017)	0.760 (0.043)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.353 (0.026)	0.439 (0.025)	0.242	0.945	0.660 (0.047)	0.190 (0.039)
LM-PP	-0.030 (0.035)	0.348 (0.021)	0.344	1.345	0.970 (0.017)	0.370 (0.048)
hier. LM	-0.063 (0.033)	0.338 (0.021)	0.323	1.267	0.920 (0.027)	0.360 (0.048)
PSCL	-0.230 (0.029)	0.370 (0.022)	0.269	1.055	0.850 (0.036)	0.210 (0.041)
MAP (100)	-0.160 (0.034)	0.378 (0.024)	0.322	1.263	0.920 (0.027)	0.270 (0.044)
MAP (75)	-0.144 (0.034)	0.372 (0.024)	0.324	1.275	0.940 (0.024)	0.280 (0.045)
MAP (50)	-0.122 (0.035)	0.366 (0.024)	0.329	1.289	0.920 (0.027)	0.290 (0.045)
<i>Mean difference: Sc5, X₅, $\delta = 2$</i>						
lrcBART: 100	-0.013 (0.020)	0.201 (0.013)	0.190	0.741	0.940 (0.024)	0.820 (0.038)
lrcBART: 75	-0.007 (0.020)	0.198 (0.013)	0.190	0.744	0.930 (0.026)	0.840 (0.037)
lrcBART: 50	-0.013 (0.021)	0.209 (0.014)	0.192	0.751	0.950 (0.022)	0.800 (0.040)
lrcBART: f90	-0.011 (0.021)	0.205 (0.013)	0.191	0.747	0.920 (0.027)	0.800 (0.040)
lrcBART: f50	-0.015 (0.020)	0.201 (0.012)	0.190	0.744	0.960 (0.020)	0.790 (0.041)
lrcBART: f25	-0.008 (0.020)	0.200 (0.012)	0.190	0.739	0.950 (0.022)	0.830 (0.038)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.679 (0.014)	0.694 (0.014)	0.150	0.585	0.010 (0.010)	0.000 (0.000)
BART-PP	-0.012 (0.020)	0.198 (0.012)	0.214	0.833	0.970 (0.017)	0.760 (0.043)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.722 (0.026)	0.767 (0.026)	0.245	0.958	0.180 (0.038)	0.010 (0.010)
LM-PP	-0.049 (0.035)	0.348 (0.020)	0.350	1.366	0.950 (0.022)	0.360 (0.048)
hier. LM	-0.105 (0.034)	0.358 (0.021)	0.330	1.290	0.930 (0.026)	0.290 (0.045)
PSCL	-0.479 (0.029)	0.560 (0.026)	0.273	1.068	0.520 (0.050)	0.050 (0.022)
MAP (100)	-0.216 (0.040)	0.454 (0.025)	0.359	1.402	0.860 (0.035)	0.180 (0.038)
MAP (75)	-0.190 (0.040)	0.439 (0.025)	0.358	1.395	0.900 (0.030)	0.210 (0.041)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
MAP (50)	-0.163 (0.039)	0.423 (0.025)	0.357	1.392	0.910 (0.029)	0.220 (0.041)
<i>Mean difference: $Sc5$, X_7, $\delta = 1$</i>						
lrcBART: 100	-0.047 (0.021)	0.217 (0.012)	0.196	0.762	0.920 (0.027)	0.720 (0.045)
lrcBART: 75	-0.041 (0.021)	0.213 (0.013)	0.194	0.752	0.920 (0.027)	0.710 (0.045)
lrcBART: 50	-0.040 (0.020)	0.208 (0.013)	0.194	0.753	0.940 (0.024)	0.780 (0.041)
lrcBART: f90	-0.047 (0.021)	0.219 (0.013)	0.197	0.767	0.930 (0.026)	0.720 (0.045)
lrcBART: f50	-0.037 (0.021)	0.214 (0.013)	0.194	0.757	0.930 (0.026)	0.750 (0.043)
lrcBART: f25	-0.028 (0.020)	0.202 (0.013)	0.192	0.754	0.940 (0.024)	0.790 (0.041)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.336 (0.015)	0.369 (0.014)	0.147	0.574	0.370 (0.048)	0.320 (0.047)
BART-PP	-0.003 (0.020)	0.198 (0.013)	0.213	0.831	0.960 (0.020)	0.770 (0.042)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.352 (0.026)	0.440 (0.025)	0.242	0.947	0.690 (0.046)	0.170 (0.038)
LM-PP	-0.025 (0.035)	0.349 (0.022)	0.342	1.342	0.940 (0.024)	0.360 (0.048)
hier. LM	-0.063 (0.033)	0.338 (0.021)	0.324	1.267	0.920 (0.027)	0.330 (0.047)
PSCL	-0.230 (0.029)	0.369 (0.023)	0.268	1.050	0.860 (0.035)	0.220 (0.041)
MAP (100)	-0.165 (0.034)	0.376 (0.024)	0.321	1.255	0.910 (0.029)	0.270 (0.044)
MAP (75)	-0.148 (0.034)	0.373 (0.024)	0.324	1.269	0.930 (0.026)	0.270 (0.044)
MAP (50)	-0.126 (0.035)	0.369 (0.024)	0.329	1.286	0.920 (0.027)	0.290 (0.045)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	-0.006 (0.019)	0.193 (0.013)	0.188	0.735	0.950 (0.022)	0.830 (0.038)
lrcBART: 75	-0.002 (0.020)	0.196 (0.013)	0.189	0.738	0.930 (0.026)	0.860 (0.035)
lrcBART: 50	-0.007 (0.020)	0.198 (0.013)	0.190	0.743	0.930 (0.026)	0.830 (0.038)
lrcBART: f90	-0.003 (0.020)	0.194 (0.013)	0.189	0.739	0.950 (0.022)	0.830 (0.038)
lrcBART: f50	-0.003 (0.020)	0.194 (0.013)	0.190	0.743	0.940 (0.024)	0.840 (0.037)
lrcBART: f25	0.002 (0.020)	0.201 (0.014)	0.191	0.743	0.920 (0.027)	0.840 (0.037)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.683 (0.016)	0.700 (0.015)	0.150	0.587	0.010 (0.010)	0.000 (0.000)
BART-PP	-0.006 (0.020)	0.196 (0.012)	0.214	0.835	0.970 (0.017)	0.770 (0.042)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.718 (0.027)	0.766 (0.027)	0.245	0.955	0.190 (0.039)	0.000 (0.000)
LM-PP	-0.046 (0.035)	0.352 (0.022)	0.349	1.363	0.960 (0.020)	0.350 (0.048)
hier. LM	-0.103 (0.034)	0.357 (0.021)	0.330	1.292	0.910 (0.029)	0.300 (0.046)
PSCL	-0.478 (0.029)	0.559 (0.026)	0.270	1.059	0.560 (0.050)	0.070 (0.026)
MAP (100)	-0.220 (0.040)	0.457 (0.025)	0.360	1.405	0.860 (0.035)	0.170 (0.038)
MAP (75)	-0.202 (0.040)	0.445 (0.024)	0.358	1.395	0.870 (0.034)	0.190 (0.039)
MAP (50)	-0.171 (0.039)	0.428 (0.025)	0.358	1.395	0.900 (0.030)	0.210 (0.041)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $IG(3/2, 3s_\tau^2/2)$. The hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.25$.

E.7 Survival two-arm results

The primary median-survival ratio is followed by the supplementary three-year RMST ratio, using the modified functional in Appendix D.5.

Table S6: Complete Survival simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: Sc1</i>						
lrcBART: 100	0.014 (0.029)	0.290 (0.037)	0.308	1.199	0.970 (0.017)	0.420 (0.049)
lrcBART: 75	0.018 (0.030)	0.295 (0.036)	0.309	1.207	0.980 (0.014)	0.440 (0.050)
lrcBART: 50	0.023 (0.031)	0.310 (0.037)	0.314	1.220	0.970 (0.017)	0.430 (0.050)
lrcBART: f90	0.014 (0.029)	0.289 (0.041)	0.300	1.175	0.970 (0.017)	0.450 (0.050)
lrcBART: f50	0.013 (0.029)	0.292 (0.038)	0.311	1.209	0.960 (0.020)	0.430 (0.050)
lrcBART: f25	0.017 (0.031)	0.308 (0.038)	0.315	1.227	0.970 (0.017)	0.410 (0.049)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	0.010 (0.022)	0.218 (0.016)	0.228	0.889	0.950 (0.022)	0.660 (0.047)
BART-PP	-0.034 (0.032)	0.316 (0.032)	0.357	1.387	0.980 (0.014)	0.300 (0.046)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	0.027 (0.031)	0.307 (0.021)	0.393	1.522	1.000 (0.000)	0.330 (0.047)
AFT-PP	0.022 (0.042)	0.415 (0.038)	0.620	2.389	0.990 (0.010)	0.100 (0.030)
hier. AFT	0.050 (0.044)	0.443 (0.060)	0.600	2.314	0.990 (0.010)	0.140 (0.035)
<i>Median-survival ratio: Sc2</i>						
lrcBART: 100	0.058 (0.034)	0.341 (0.030)	0.325	1.261	0.960 (0.020)	0.480 (0.050)
lrcBART: 75	0.061 (0.034)	0.346 (0.032)	0.326	1.268	0.940 (0.024)	0.430 (0.050)
lrcBART: 50	0.057 (0.034)	0.347 (0.034)	0.324	1.265	0.960 (0.020)	0.470 (0.050)
lrcBART: f90	0.058 (0.034)	0.345 (0.031)	0.326	1.267	0.950 (0.022)	0.480 (0.050)
lrcBART: f50	0.047 (0.035)	0.348 (0.031)	0.329	1.279	0.960 (0.020)	0.460 (0.050)
lrcBART: f25	0.042 (0.036)	0.357 (0.035)	0.331	1.287	0.960 (0.020)	0.420 (0.049)
BART-NP	0.117 (0.043)	0.440 (0.060)	0.444	1.725	0.960 (0.020)	0.320 (0.047)
BART-CP	0.130 (0.028)	0.310 (0.028)	0.330	1.288	0.970 (0.017)	0.630 (0.048)
BART-PP	-0.019 (0.035)	0.349 (0.025)	0.370	1.431	0.970 (0.017)	0.280 (0.045)
AFT-NP	0.018 (0.053)	0.529 (0.049)	0.593	2.280	0.960 (0.020)	0.160 (0.037)
AFT-CP	1.202 (0.079)	1.435 (0.093)	0.906	3.518	0.570 (0.050)	0.920 (0.027)
AFT-PP	0.403 (0.071)	0.817 (0.083)	0.785	3.014	0.920 (0.027)	0.300 (0.046)
hier. AFT	0.417 (0.070)	0.815 (0.078)	0.765	2.944	0.930 (0.026)	0.390 (0.049)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: Sc3, $p = -0.5$</i>						
lrcBART: 100	0.041 (0.053)	0.531 (0.074)	0.488	1.890	0.920 (0.027)	0.290 (0.045)
lrcBART: 75	0.044 (0.054)	0.535 (0.076)	0.486	1.873	0.920 (0.027)	0.330 (0.047)
lrcBART: 50	0.047 (0.053)	0.532 (0.077)	0.488	1.895	0.930 (0.026)	0.290 (0.045)
lrcBART: f90	0.043 (0.054)	0.542 (0.082)	0.489	1.885	0.910 (0.029)	0.280 (0.045)
lrcBART: f50	0.042 (0.053)	0.526 (0.075)	0.491	1.883	0.930 (0.026)	0.290 (0.045)
lrcBART: f25	0.044 (0.053)	0.524 (0.073)	0.489	1.890	0.930 (0.026)	0.310 (0.046)
BART-NP	0.036 (0.057)	0.566 (0.075)	0.548	2.114	0.930 (0.026)	0.210 (0.041)
BART-CP	2.120 (0.069)	2.228 (0.071)	0.790	3.069	0.000 (0.000)	1.000 (0.000)
BART-PP	0.029 (0.047)	0.472 (0.059)	0.476	1.836	0.930 (0.026)	0.240 (0.043)
AFT-NP	-0.028 (0.060)	0.601 (0.073)	0.633	2.433	0.940 (0.024)	0.150 (0.036)
AFT-CP	2.647 (0.107)	2.852 (0.116)	1.231	4.752	0.050 (0.022)	1.000 (0.000)
AFT-PP	0.463 (0.077)	0.894 (0.092)	0.831	3.188	0.920 (0.027)	0.390 (0.049)
hier. AFT	0.445 (0.082)	0.933 (0.102)	0.839	3.217	0.910 (0.029)	0.370 (0.048)
<i>Median-survival ratio: Sc3, $p = 0$</i>						
lrcBART: 100	0.076 (0.047)	0.472 (0.037)	0.521	2.012	0.970 (0.017)	0.290 (0.045)
lrcBART: 75	0.083 (0.048)	0.482 (0.038)	0.520	2.010	0.960 (0.020)	0.310 (0.046)
lrcBART: 50	0.081 (0.047)	0.479 (0.038)	0.522	2.018	0.970 (0.017)	0.290 (0.045)
lrcBART: f90	0.071 (0.047)	0.471 (0.038)	0.521	2.019	0.930 (0.026)	0.300 (0.046)
lrcBART: f50	0.079 (0.047)	0.471 (0.037)	0.521	2.003	0.970 (0.017)	0.290 (0.045)
lrcBART: f25	0.080 (0.047)	0.475 (0.038)	0.521	2.015	0.960 (0.020)	0.280 (0.045)
BART-NP	0.014 (0.050)	0.502 (0.037)	0.569	2.190	0.960 (0.020)	0.180 (0.038)
BART-CP	2.642 (0.094)	2.802 (0.104)	0.885	3.439	0.000 (0.000)	1.000 (0.000)
BART-PP	0.075 (0.044)	0.446 (0.033)	0.503	1.940	0.960 (0.020)	0.300 (0.046)
AFT-NP	-0.090 (0.053)	0.531 (0.034)	0.632	2.410	0.970 (0.017)	0.110 (0.031)
AFT-CP	3.187 (0.127)	3.429 (0.141)	1.331	5.143	0.010 (0.010)	1.000 (0.000)
AFT-PP	0.477 (0.069)	0.839 (0.064)	0.844	3.220	0.930 (0.026)	0.470 (0.050)
hier. AFT	0.415 (0.073)	0.836 (0.066)	0.870	3.330	0.920 (0.027)	0.350 (0.048)
<i>Median-survival ratio: Sc3, $p = 0.5$</i>						
lrcBART: 100	0.004 (0.047)	0.468 (0.040)	0.483	1.863	0.930 (0.026)	0.240 (0.043)
lrcBART: 75	0.014 (0.048)	0.475 (0.040)	0.477	1.853	0.930 (0.026)	0.260 (0.044)
lrcBART: 50	0.016 (0.048)	0.477 (0.040)	0.483	1.864	0.930 (0.026)	0.250 (0.043)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: f90	0.009 (0.047)	0.469 (0.041)	0.481	1.860	0.930 (0.026)	0.270 (0.044)
lrcBART: f50	0.012 (0.047)	0.469 (0.040)	0.484	1.865	0.930 (0.026)	0.260 (0.044)
lrcBART: f25	0.012 (0.047)	0.469 (0.039)	0.481	1.860	0.930 (0.026)	0.290 (0.045)
BART-NP	-0.004 (0.052)	0.514 (0.036)	0.540	2.090	0.940 (0.024)	0.200 (0.040)
BART-CP	2.083 (0.079)	2.225 (0.084)	0.775	3.013	0.030 (0.017)	1.000 (0.000)
BART-PP	0.006 (0.045)	0.445 (0.034)	0.470	1.815	0.940 (0.024)	0.250 (0.043)
AFT-NP	-0.083 (0.054)	0.544 (0.037)	0.616	2.352	0.910 (0.029)	0.120 (0.032)
AFT-CP	2.601 (0.124)	2.878 (0.133)	1.220	4.716	0.070 (0.026)	1.000 (0.000)
AFT-PP	0.422 (0.072)	0.830 (0.065)	0.827	3.164	0.950 (0.022)	0.400 (0.049)
hier. AFT	0.391 (0.076)	0.849 (0.063)	0.830	3.184	0.930 (0.026)	0.350 (0.048)
<i>Median-survival ratio: Sc4, $\delta = 1$</i>						
lrcBART: 100	-0.007 (0.033)	0.332 (0.040)	0.329	1.277	0.940 (0.024)	0.370 (0.048)
lrcBART: 75	-0.005 (0.033)	0.332 (0.037)	0.330	1.279	0.960 (0.020)	0.370 (0.048)
lrcBART: 50	-0.013 (0.033)	0.331 (0.041)	0.326	1.264	0.950 (0.022)	0.380 (0.049)
lrcBART: f90	-0.006 (0.034)	0.335 (0.040)	0.327	1.271	0.950 (0.022)	0.400 (0.049)
lrcBART: f50	-0.007 (0.034)	0.335 (0.041)	0.328	1.279	0.940 (0.024)	0.370 (0.048)
lrcBART: f25	-0.013 (0.033)	0.332 (0.039)	0.331	1.287	0.940 (0.024)	0.360 (0.048)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.729 (0.010)	0.736 (0.009)	0.112	0.436	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.080 (0.030)	0.311 (0.029)	0.346	1.342	0.960 (0.020)	0.230 (0.042)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.768 (0.013)	0.780 (0.013)	0.180	0.697	0.080 (0.027)	0.000 (0.000)
AFT-PP	-0.212 (0.035)	0.405 (0.024)	0.526	2.015	0.960 (0.020)	0.030 (0.017)
hier. AFT	-0.159 (0.043)	0.459 (0.051)	0.545	2.094	0.970 (0.017)	0.060 (0.024)
<i>Median-survival ratio: Sc4, $\delta = 2$</i>						
lrcBART: 100	-0.005 (0.033)	0.325 (0.041)	0.326	1.272	0.960 (0.020)	0.340 (0.047)
lrcBART: 75	-0.006 (0.033)	0.327 (0.042)	0.326	1.265	0.960 (0.020)	0.350 (0.048)
lrcBART: 50	-0.002 (0.033)	0.330 (0.040)	0.329	1.281	0.970 (0.017)	0.400 (0.049)
lrcBART: f90	-0.007 (0.033)	0.327 (0.040)	0.329	1.277	0.970 (0.017)	0.360 (0.048)
lrcBART: f50	-0.006 (0.033)	0.331 (0.042)	0.328	1.274	0.960 (0.020)	0.380 (0.049)
lrcBART: f25	-0.002 (0.033)	0.327 (0.042)	0.329	1.282	0.980 (0.014)	0.370 (0.048)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
BART-CP	-1.075 (0.005)	1.076 (0.005)	0.058	0.227	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.098 (0.030)	0.317 (0.024)	0.340	1.315	0.960 (0.020)	0.230 (0.042)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-1.123 (0.006)	1.124 (0.006)	0.083	0.322	0.000 (0.000)	0.000 (0.000)
AFT-PP	-0.434 (0.030)	0.524 (0.023)	0.430	1.643	0.890 (0.031)	0.010 (0.010)
hier. AFT	-0.187 (0.047)	0.502 (0.050)	0.560	2.145	0.950 (0.022)	0.050 (0.022)
<i>Median-survival ratio: Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	-0.033 (0.031)	0.313 (0.040)	0.312	1.210	0.950 (0.022)	0.310 (0.046)
lrcBART: 75	-0.020 (0.032)	0.316 (0.042)	0.311	1.202	0.960 (0.020)	0.370 (0.048)
lrcBART: 50	-0.019 (0.032)	0.320 (0.047)	0.316	1.228	0.940 (0.024)	0.330 (0.047)
lrcBART: f90	-0.043 (0.030)	0.306 (0.039)	0.304	1.179	0.960 (0.020)	0.350 (0.048)
lrcBART: f50	-0.026 (0.031)	0.306 (0.039)	0.308	1.193	0.950 (0.022)	0.350 (0.048)
lrcBART: f25	-0.017 (0.031)	0.310 (0.037)	0.314	1.221	0.970 (0.017)	0.390 (0.049)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.209 (0.019)	0.279 (0.013)	0.194	0.756	0.840 (0.037)	0.260 (0.044)
BART-PP	-0.050 (0.031)	0.310 (0.030)	0.352	1.370	0.970 (0.017)	0.290 (0.045)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.228 (0.025)	0.338 (0.018)	0.326	1.260	0.890 (0.031)	0.110 (0.031)
AFT-PP	-0.040 (0.040)	0.402 (0.033)	0.597	2.288	1.000 (0.000)	0.100 (0.030)
hier. AFT	-0.013 (0.044)	0.434 (0.059)	0.575	2.206	0.980 (0.014)	0.100 (0.030)
<i>Median-survival ratio: Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	-0.054 (0.033)	0.331 (0.039)	0.320	1.245	0.930 (0.026)	0.330 (0.047)
lrcBART: 75	-0.044 (0.034)	0.339 (0.043)	0.321	1.244	0.940 (0.024)	0.330 (0.047)
lrcBART: 50	-0.036 (0.033)	0.329 (0.040)	0.322	1.247	0.950 (0.022)	0.330 (0.047)
lrcBART: f90	-0.057 (0.034)	0.341 (0.040)	0.321	1.247	0.930 (0.026)	0.330 (0.047)
lrcBART: f50	-0.051 (0.032)	0.327 (0.041)	0.320	1.241	0.940 (0.024)	0.320 (0.047)
lrcBART: f25	-0.034 (0.033)	0.335 (0.039)	0.324	1.259	0.950 (0.022)	0.330 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.391 (0.016)	0.422 (0.014)	0.165	0.645	0.420 (0.049)	0.070 (0.026)
BART-PP	-0.067 (0.031)	0.316 (0.031)	0.352	1.367	0.960 (0.020)	0.240 (0.043)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.441 (0.020)	0.484 (0.018)	0.269	1.040	0.700 (0.046)	0.000 (0.000)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
AFT-PP	-0.116 (0.037)	0.385 (0.026)	0.573	2.193	1.000 (0.000)	0.060 (0.024)
hier. AFT	-0.074 (0.043)	0.433 (0.056)	0.562	2.160	0.980 (0.014)	0.080 (0.027)
<i>Median-survival ratio: Sc5, X₅, $\delta = 2$</i>						
lrcBART: 100	-0.034 (0.035)	0.348 (0.045)	0.333	1.293	0.970 (0.017)	0.340 (0.047)
lrcBART: 75	-0.035 (0.035)	0.348 (0.044)	0.334	1.294	0.950 (0.022)	0.340 (0.047)
lrcBART: 50	-0.036 (0.035)	0.347 (0.048)	0.332	1.296	0.950 (0.022)	0.330 (0.047)
lrcBART: f90	-0.039 (0.035)	0.347 (0.040)	0.333	1.294	0.950 (0.022)	0.310 (0.046)
lrcBART: f50	-0.043 (0.034)	0.342 (0.045)	0.331	1.288	0.960 (0.020)	0.340 (0.047)
lrcBART: f25	-0.042 (0.034)	0.344 (0.042)	0.328	1.277	0.940 (0.024)	0.330 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.659 (0.012)	0.670 (0.011)	0.126	0.494	0.010 (0.010)	0.000 (0.000)
BART-PP	-0.086 (0.032)	0.326 (0.032)	0.349	1.357	0.970 (0.017)	0.230 (0.042)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.741 (0.014)	0.754 (0.014)	0.190	0.738	0.170 (0.038)	0.000 (0.000)
AFT-PP	-0.244 (0.033)	0.413 (0.023)	0.521	1.993	0.950 (0.022)	0.040 (0.020)
hier. AFT	-0.156 (0.043)	0.457 (0.050)	0.545	2.084	0.970 (0.017)	0.050 (0.022)
<i>Median-survival ratio: Sc5, X₇, $\delta = 1$</i>						
lrcBART: 100	-0.041 (0.034)	0.338 (0.037)	0.323	1.256	0.940 (0.024)	0.300 (0.046)
lrcBART: 75	-0.028 (0.035)	0.345 (0.038)	0.327	1.264	0.940 (0.024)	0.330 (0.047)
lrcBART: 50	-0.031 (0.033)	0.333 (0.039)	0.325	1.262	0.960 (0.020)	0.330 (0.047)
lrcBART: f90	-0.052 (0.034)	0.340 (0.032)	0.325	1.262	0.940 (0.024)	0.300 (0.046)
lrcBART: f50	-0.034 (0.034)	0.339 (0.037)	0.326	1.269	0.920 (0.027)	0.310 (0.046)
lrcBART: f25	-0.031 (0.033)	0.331 (0.036)	0.322	1.255	0.970 (0.017)	0.330 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.428 (0.014)	0.451 (0.012)	0.160	0.623	0.340 (0.047)	0.030 (0.017)
BART-PP	-0.064 (0.032)	0.320 (0.032)	0.353	1.370	0.970 (0.017)	0.260 (0.044)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.454 (0.020)	0.495 (0.018)	0.265	1.029	0.700 (0.046)	0.000 (0.000)
AFT-PP	-0.101 (0.038)	0.395 (0.029)	0.571	2.188	1.000 (0.000)	0.080 (0.027)
hier. AFT	-0.073 (0.043)	0.430 (0.055)	0.559	2.146	0.980 (0.014)	0.100 (0.030)
<i>Median-survival ratio: Sc5, X₇, $\delta = 2$</i>						
lrcBART: 100	-0.016 (0.033)	0.331 (0.033)	0.332	1.288	0.930 (0.026)	0.340 (0.047)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: 75	-0.018 (0.033)	0.333 (0.038)	0.330	1.281	0.960 (0.020)	0.350 (0.048)
lrcBART: 50	-0.017 (0.033)	0.330 (0.036)	0.331	1.288	0.950 (0.022)	0.350 (0.048)
lrcBART: f90	-0.010 (0.034)	0.339 (0.039)	0.330	1.285	0.950 (0.022)	0.370 (0.048)
lrcBART: f50	-0.018 (0.034)	0.338 (0.041)	0.330	1.280	0.940 (0.024)	0.350 (0.048)
lrcBART: f25	-0.020 (0.035)	0.346 (0.041)	0.330	1.282	0.970 (0.017)	0.340 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.718 (0.010)	0.725 (0.010)	0.115	0.450	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.083 (0.031)	0.318 (0.028)	0.352	1.371	0.970 (0.017)	0.240 (0.043)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.769 (0.013)	0.780 (0.013)	0.182	0.705	0.110 (0.031)	0.000 (0.000)
AFT-PP	-0.234 (0.035)	0.424 (0.025)	0.521	1.995	0.940 (0.024)	0.030 (0.017)
hier. AFT	-0.158 (0.044)	0.461 (0.052)	0.547	2.098	0.960 (0.020)	0.060 (0.024)
<i>Modified RMST ratio: Sc1</i>						
lrcBART: 100	0.003 (0.009)	0.087 (0.007)	0.097	0.378	0.960 (0.020)	0.340 (0.047)
lrcBART: 75	0.004 (0.009)	0.088 (0.007)	0.097	0.379	0.970 (0.017)	0.360 (0.048)
lrcBART: 50	0.005 (0.009)	0.092 (0.007)	0.098	0.384	0.960 (0.020)	0.370 (0.048)
lrcBART: f90	0.003 (0.009)	0.087 (0.008)	0.094	0.371	0.970 (0.017)	0.390 (0.049)
lrcBART: f50	0.001 (0.009)	0.088 (0.008)	0.097	0.379	0.960 (0.020)	0.340 (0.047)
lrcBART: f25	0.003 (0.009)	0.091 (0.008)	0.099	0.386	0.960 (0.020)	0.350 (0.048)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.004 (0.007)	0.071 (0.005)	0.072	0.280	0.950 (0.022)	0.520 (0.050)
BART-PP	-0.016 (0.010)	0.096 (0.007)	0.111	0.432	0.980 (0.014)	0.240 (0.043)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.011 (0.008)	0.078 (0.005)	0.090	0.352	0.970 (0.017)	0.350 (0.048)
AFT-PP	-0.015 (0.010)	0.103 (0.007)	0.140	0.548	1.000 (0.000)	0.100 (0.030)
hier. AFT	-0.002 (0.011)	0.110 (0.010)	0.143	0.562	0.990 (0.010)	0.130 (0.034)
<i>Modified RMST ratio: Sc2</i>						
lrcBART: 100	0.026 (0.010)	0.107 (0.010)	0.097	0.378	0.930 (0.026)	0.470 (0.050)
lrcBART: 75	0.027 (0.011)	0.108 (0.010)	0.097	0.377	0.940 (0.024)	0.460 (0.050)
lrcBART: 50	0.026 (0.011)	0.110 (0.011)	0.096	0.379	0.940 (0.024)	0.470 (0.050)
lrcBART: f90	0.026 (0.010)	0.107 (0.009)	0.097	0.377	0.930 (0.026)	0.500 (0.050)
lrcBART: f50	0.023 (0.011)	0.109 (0.010)	0.097	0.380	0.930 (0.026)	0.450 (0.050)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: f25	0.020 (0.011)	0.109 (0.011)	0.097	0.382	0.940 (0.024)	0.410 (0.049)
BART-NP	0.093 (0.015)	0.174 (0.018)	0.153	0.598	0.910 (0.029)	0.460 (0.050)
BART-CP	0.033 (0.009)	0.094 (0.009)	0.096	0.374	0.960 (0.020)	0.550 (0.050)
BART-PP	-0.005 (0.011)	0.109 (0.008)	0.107	0.419	0.940 (0.024)	0.280 (0.045)
AFT-NP	-0.012 (0.012)	0.122 (0.009)	0.132	0.516	0.980 (0.014)	0.170 (0.038)
AFT-CP	0.278 (0.014)	0.313 (0.016)	0.157	0.614	0.460 (0.050)	0.970 (0.017)
AFT-PP	0.101 (0.015)	0.184 (0.016)	0.161	0.626	0.880 (0.032)	0.440 (0.050)
hier. AFT	0.084 (0.015)	0.168 (0.015)	0.153	0.597	0.930 (0.026)	0.410 (0.049)
<i>Modified RMST ratio: Sc3, $\rho = -0.5$</i>						
lrcBART: 100	0.014 (0.015)	0.152 (0.019)	0.135	0.528	0.930 (0.026)	0.250 (0.043)
lrcBART: 75	0.014 (0.015)	0.151 (0.018)	0.134	0.521	0.920 (0.027)	0.300 (0.046)
lrcBART: 50	0.015 (0.015)	0.152 (0.020)	0.135	0.527	0.920 (0.027)	0.250 (0.043)
lrcBART: f90	0.014 (0.015)	0.152 (0.019)	0.135	0.527	0.920 (0.027)	0.250 (0.043)
lrcBART: f50	0.014 (0.015)	0.149 (0.019)	0.135	0.527	0.940 (0.024)	0.270 (0.044)
lrcBART: f25	0.015 (0.015)	0.149 (0.017)	0.135	0.527	0.940 (0.024)	0.260 (0.044)
BART-NP	0.013 (0.016)	0.157 (0.015)	0.152	0.595	0.950 (0.022)	0.230 (0.042)
BART-CP	0.490 (0.014)	0.509 (0.015)	0.146	0.572	0.000 (0.000)	1.000 (0.000)
BART-PP	0.006 (0.013)	0.131 (0.014)	0.128	0.501	0.930 (0.026)	0.260 (0.044)
AFT-NP	-0.035 (0.013)	0.135 (0.011)	0.135	0.527	0.920 (0.027)	0.130 (0.034)
AFT-CP	0.480 (0.016)	0.504 (0.016)	0.156	0.609	0.060 (0.024)	1.000 (0.000)
AFT-PP	0.092 (0.016)	0.186 (0.016)	0.155	0.607	0.930 (0.026)	0.450 (0.050)
hier. AFT	0.078 (0.016)	0.180 (0.016)	0.160	0.624	0.940 (0.024)	0.370 (0.048)
<i>Modified RMST ratio: Sc3, $\rho = 0$</i>						
lrcBART: 100	0.021 (0.013)	0.136 (0.009)	0.138	0.538	0.960 (0.020)	0.290 (0.045)
lrcBART: 75	0.024 (0.014)	0.139 (0.009)	0.138	0.541	0.960 (0.020)	0.300 (0.046)
lrcBART: 50	0.023 (0.014)	0.138 (0.009)	0.138	0.540	0.970 (0.017)	0.300 (0.046)
lrcBART: f90	0.020 (0.014)	0.136 (0.009)	0.137	0.537	0.970 (0.017)	0.280 (0.045)
lrcBART: f50	0.022 (0.013)	0.136 (0.009)	0.138	0.537	0.960 (0.020)	0.290 (0.045)
lrcBART: f25	0.023 (0.014)	0.137 (0.009)	0.138	0.539	0.960 (0.020)	0.290 (0.045)
BART-NP	0.006 (0.015)	0.146 (0.009)	0.155	0.606	0.980 (0.014)	0.180 (0.038)
BART-CP	0.583 (0.016)	0.604 (0.016)	0.148	0.579	0.000 (0.000)	1.000 (0.000)
BART-PP	0.019 (0.012)	0.122 (0.007)	0.131	0.508	0.980 (0.014)	0.320 (0.047)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
AFT-NP	-0.054 (0.012)	0.134 (0.007)	0.132	0.516	0.950 (0.022)	0.110 (0.031)
AFT-CP	0.549 (0.016)	0.570 (0.016)	0.153	0.598	0.010 (0.010)	1.000 (0.000)
AFT-PP	0.092 (0.014)	0.166 (0.011)	0.155	0.604	0.930 (0.026)	0.510 (0.050)
hier. AFT	0.062 (0.015)	0.161 (0.011)	0.158	0.614	0.940 (0.024)	0.360 (0.048)
<i>Modified RMST ratio: Sc3, $\rho = 0.5$</i>						
lrcBART: 100	-0.001 (0.014)	0.134 (0.011)	0.133	0.518	0.920 (0.027)	0.240 (0.043)
lrcBART: 75	0.003 (0.014)	0.136 (0.010)	0.132	0.516	0.930 (0.026)	0.260 (0.044)
lrcBART: 50	0.003 (0.014)	0.137 (0.010)	0.133	0.518	0.920 (0.027)	0.290 (0.045)
lrcBART: f90	0.002 (0.014)	0.135 (0.011)	0.133	0.517	0.930 (0.026)	0.260 (0.044)
lrcBART: f50	0.002 (0.014)	0.135 (0.010)	0.133	0.517	0.920 (0.027)	0.240 (0.043)
lrcBART: f25	0.002 (0.014)	0.135 (0.010)	0.133	0.519	0.940 (0.024)	0.270 (0.044)
BART-NP	0.004 (0.016)	0.155 (0.012)	0.153	0.601	0.970 (0.017)	0.200 (0.040)
BART-CP	0.478 (0.014)	0.499 (0.014)	0.143	0.563	0.030 (0.017)	1.000 (0.000)
BART-PP	-0.004 (0.012)	0.124 (0.008)	0.127	0.496	0.940 (0.024)	0.250 (0.043)
AFT-NP	-0.052 (0.013)	0.141 (0.010)	0.132	0.517	0.900 (0.030)	0.130 (0.034)
AFT-CP	0.458 (0.016)	0.486 (0.016)	0.153	0.597	0.070 (0.026)	1.000 (0.000)
AFT-PP	0.076 (0.015)	0.166 (0.011)	0.154	0.600	0.940 (0.024)	0.440 (0.050)
hier. AFT	0.060 (0.016)	0.172 (0.013)	0.157	0.615	0.940 (0.024)	0.380 (0.049)
<i>Modified RMST ratio: Sc4, $\delta = 1$</i>						
lrcBART: 100	-0.001 (0.010)	0.101 (0.008)	0.105	0.410	0.960 (0.020)	0.320 (0.047)
lrcBART: 75	-0.001 (0.010)	0.099 (0.008)	0.105	0.411	0.960 (0.020)	0.300 (0.046)
lrcBART: 50	-0.003 (0.010)	0.100 (0.008)	0.105	0.410	0.980 (0.014)	0.290 (0.045)
lrcBART: f90	-0.001 (0.010)	0.101 (0.008)	0.105	0.410	0.970 (0.017)	0.310 (0.046)
lrcBART: f50	-0.001 (0.010)	0.100 (0.008)	0.105	0.411	0.970 (0.017)	0.290 (0.045)
lrcBART: f25	-0.002 (0.010)	0.101 (0.008)	0.106	0.414	0.960 (0.020)	0.300 (0.046)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.259 (0.005)	0.263 (0.005)	0.051	0.200	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.023 (0.009)	0.096 (0.007)	0.112	0.437	0.980 (0.014)	0.180 (0.038)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.249 (0.006)	0.255 (0.006)	0.068	0.266	0.030 (0.017)	0.000 (0.000)
AFT-PP	-0.076 (0.009)	0.120 (0.007)	0.129	0.506	0.940 (0.024)	0.040 (0.020)
hier. AFT	-0.059 (0.011)	0.128 (0.009)	0.139	0.545	0.970 (0.017)	0.060 (0.024)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Modified RMST ratio: Sc4, $\delta = 2$</i>						
lrcBART: 100	0.008 (0.010)	0.100 (0.009)	0.108	0.422	0.960 (0.020)	0.300 (0.046)
lrcBART: 75	0.008 (0.010)	0.102 (0.009)	0.108	0.417	0.960 (0.020)	0.300 (0.046)
lrcBART: 50	0.009 (0.010)	0.102 (0.008)	0.108	0.423	0.970 (0.017)	0.320 (0.047)
lrcBART: f90	0.007 (0.010)	0.101 (0.009)	0.108	0.420	0.950 (0.022)	0.320 (0.047)
lrcBART: f50	0.008 (0.010)	0.102 (0.009)	0.108	0.422	0.950 (0.022)	0.310 (0.046)
lrcBART: f25	0.009 (0.010)	0.100 (0.009)	0.108	0.422	0.970 (0.017)	0.330 (0.047)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.427 (0.004)	0.429 (0.004)	0.040	0.156	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.017 (0.010)	0.099 (0.007)	0.115	0.451	0.980 (0.014)	0.220 (0.041)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.407 (0.005)	0.410 (0.005)	0.055	0.216	0.000 (0.000)	0.000 (0.000)
AFT-PP	-0.140 (0.009)	0.166 (0.008)	0.119	0.463	0.850 (0.036)	0.010 (0.010)
hier. AFT	-0.069 (0.012)	0.141 (0.009)	0.145	0.564	0.930 (0.026)	0.060 (0.024)
<i>Modified RMST ratio: Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	-0.023 (0.009)	0.092 (0.007)	0.095	0.372	0.960 (0.020)	0.240 (0.043)
lrcBART: 75	-0.019 (0.009)	0.091 (0.007)	0.095	0.370	0.960 (0.020)	0.270 (0.044)
lrcBART: 50	-0.020 (0.009)	0.092 (0.008)	0.096	0.375	0.950 (0.022)	0.270 (0.044)
lrcBART: f90	-0.027 (0.009)	0.091 (0.007)	0.093	0.364	0.940 (0.024)	0.230 (0.042)
lrcBART: f50	-0.021 (0.009)	0.089 (0.007)	0.094	0.369	0.970 (0.017)	0.260 (0.044)
lrcBART: f25	-0.019 (0.009)	0.092 (0.007)	0.096	0.377	0.970 (0.017)	0.280 (0.045)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.080 (0.006)	0.102 (0.005)	0.065	0.252	0.750 (0.043)	0.180 (0.038)
BART-PP	-0.029 (0.009)	0.094 (0.006)	0.107	0.418	0.960 (0.020)	0.210 (0.041)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.082 (0.007)	0.108 (0.006)	0.083	0.324	0.860 (0.035)	0.110 (0.031)
AFT-PP	-0.036 (0.010)	0.105 (0.006)	0.135	0.529	1.000 (0.000)	0.090 (0.029)
hier. AFT	-0.020 (0.011)	0.111 (0.010)	0.140	0.549	0.990 (0.010)	0.110 (0.031)
<i>Modified RMST ratio: Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	-0.041 (0.009)	0.100 (0.006)	0.096	0.374	0.900 (0.030)	0.200 (0.040)
lrcBART: 75	-0.039 (0.009)	0.100 (0.007)	0.096	0.373	0.920 (0.027)	0.190 (0.039)
lrcBART: 50	-0.038 (0.009)	0.098 (0.006)	0.096	0.374	0.940 (0.024)	0.190 (0.039)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: f90	-0.043 (0.009)	0.103 (0.007)	0.096	0.374	0.900 (0.030)	0.180 (0.038)
lrcBART: f50	-0.041 (0.009)	0.098 (0.006)	0.095	0.372	0.930 (0.026)	0.190 (0.039)
lrcBART: f25	-0.035 (0.009)	0.099 (0.006)	0.096	0.375	0.940 (0.024)	0.190 (0.039)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.146 (0.006)	0.157 (0.005)	0.058	0.228	0.340 (0.047)	0.030 (0.017)
BART-PP	-0.042 (0.009)	0.098 (0.007)	0.105	0.412	0.950 (0.022)	0.130 (0.034)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.145 (0.006)	0.159 (0.006)	0.077	0.299	0.570 (0.050)	0.000 (0.000)
AFT-PP	-0.060 (0.009)	0.110 (0.007)	0.131	0.510	0.990 (0.010)	0.070 (0.026)
hier. AFT	-0.037 (0.011)	0.115 (0.009)	0.139	0.543	0.990 (0.010)	0.080 (0.027)
<i>Modified RMST ratio: Sc5, X₅, $\delta = 2$</i>						
lrcBART: 100	-0.042 (0.010)	0.105 (0.007)	0.099	0.386	0.950 (0.022)	0.120 (0.032)
lrcBART: 75	-0.044 (0.009)	0.104 (0.007)	0.099	0.387	0.920 (0.027)	0.160 (0.037)
lrcBART: 50	-0.043 (0.009)	0.104 (0.007)	0.099	0.387	0.940 (0.024)	0.180 (0.038)
lrcBART: f90	-0.045 (0.009)	0.104 (0.007)	0.099	0.384	0.930 (0.026)	0.140 (0.035)
lrcBART: f50	-0.047 (0.009)	0.105 (0.007)	0.099	0.385	0.920 (0.027)	0.160 (0.037)
lrcBART: f25	-0.044 (0.009)	0.104 (0.007)	0.098	0.382	0.910 (0.029)	0.160 (0.037)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.247 (0.005)	0.252 (0.005)	0.051	0.197	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.052 (0.009)	0.102 (0.007)	0.105	0.411	0.950 (0.022)	0.110 (0.031)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.244 (0.006)	0.250 (0.006)	0.068	0.266	0.020 (0.014)	0.000 (0.000)
AFT-PP	-0.102 (0.009)	0.134 (0.007)	0.121	0.473	0.920 (0.027)	0.030 (0.017)
hier. AFT	-0.060 (0.011)	0.128 (0.009)	0.138	0.538	0.970 (0.017)	0.060 (0.024)
<i>Modified RMST ratio: Sc5, X₇, $\delta = 1$</i>						
lrcBART: 100	-0.019 (0.010)	0.100 (0.007)	0.102	0.398	0.940 (0.024)	0.250 (0.043)
lrcBART: 75	-0.015 (0.010)	0.102 (0.008)	0.102	0.397	0.930 (0.026)	0.240 (0.043)
lrcBART: 50	-0.015 (0.010)	0.099 (0.007)	0.102	0.398	0.930 (0.026)	0.250 (0.043)
lrcBART: f90	-0.022 (0.010)	0.104 (0.007)	0.102	0.399	0.930 (0.026)	0.260 (0.044)
lrcBART: f50	-0.016 (0.010)	0.102 (0.007)	0.102	0.397	0.940 (0.024)	0.270 (0.044)
lrcBART: f25	-0.016 (0.010)	0.099 (0.007)	0.101	0.395	0.940 (0.024)	0.260 (0.044)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
BART-CP	-0.148 (0.005)	0.157 (0.005)	0.060	0.234	0.360 (0.048)	0.010 (0.010)
BART-PP	-0.026 (0.009)	0.098 (0.007)	0.110	0.432	0.960 (0.020)	0.210 (0.041)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.145 (0.006)	0.158 (0.006)	0.077	0.302	0.620 (0.049)	0.000 (0.000)
AFT-PP	-0.048 (0.010)	0.110 (0.007)	0.134	0.525	0.980 (0.014)	0.080 (0.027)
hier. AFT	-0.035 (0.011)	0.115 (0.010)	0.139	0.543	0.980 (0.014)	0.090 (0.029)
<i>Modified RMST ratio: Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	-0.014 (0.010)	0.100 (0.007)	0.103	0.404	0.950 (0.022)	0.290 (0.045)
lrcBART: 75	-0.016 (0.010)	0.098 (0.008)	0.103	0.402	0.970 (0.017)	0.260 (0.044)
lrcBART: 50	-0.016 (0.010)	0.097 (0.007)	0.103	0.402	0.960 (0.020)	0.270 (0.044)
lrcBART: f90	-0.014 (0.010)	0.099 (0.008)	0.103	0.401	0.950 (0.022)	0.310 (0.046)
lrcBART: f50	-0.017 (0.010)	0.100 (0.008)	0.103	0.402	0.940 (0.024)	0.270 (0.044)
lrcBART: f25	-0.018 (0.010)	0.102 (0.008)	0.103	0.402	0.950 (0.022)	0.260 (0.044)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.259 (0.005)	0.264 (0.005)	0.051	0.200	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.031 (0.009)	0.098 (0.007)	0.112	0.438	0.950 (0.022)	0.190 (0.039)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.250 (0.006)	0.257 (0.006)	0.068	0.266	0.040 (0.020)	0.000 (0.000)
AFT-PP	-0.089 (0.010)	0.130 (0.008)	0.127	0.493	0.920 (0.027)	0.050 (0.022)
hier. AFT	-0.059 (0.011)	0.128 (0.009)	0.140	0.545	0.960 (0.020)	0.070 (0.026)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $IG(3/2, 3s_T^2/2)$. The hierarchical comparator uses discrepancy-variance prior scale $s_T^2 = 0.25$.

E.8 Gaussian single-arm results

Sc1 and Sc2 at $n_{\text{trt}} = 200$ include all five lrcBART priors and all three comparator settings.

Table S7: Complete Gaussian single arm simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	-0.015 (0.016)	0.156 (0.012)	0.176	0.689	0.950 (0.022)	0.890 (0.031)
lrcBART: f90	-0.012 (0.015)	0.153 (0.011)	0.163	0.636	0.960 (0.020)	0.920 (0.027)
lrcBART: s0min	-0.015 (0.016)	0.157 (0.012)	0.151	0.591	0.910 (0.029)	0.920 (0.027)
lrcBART: $w = 0.9$, 100	-0.010 (0.016)	0.155 (0.011)	1.289	6.036	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	-0.007 (0.016)	0.159 (0.010)	1.292	6.067	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.014 (0.015)	0.153 (0.011)	0.150	0.588	0.940 (0.024)	0.940 (0.024)
LM-CP	-0.029 (0.027)	0.270 (0.017)	0.252	0.983	0.960 (0.020)	0.600 (0.049)
hier. LM	-0.025 (0.027)	0.272 (0.017)	0.576	2.210	1.000 (0.000)	0.090 (0.029)
<i>Mean difference: $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	0.062 (0.017)	0.177 (0.010)	0.179	0.697	0.980 (0.014)	0.910 (0.029)
lrcBART: f90	0.064 (0.017)	0.183 (0.010)	0.184	0.714	0.970 (0.017)	0.910 (0.029)
lrcBART: s0min	0.061 (0.017)	0.177 (0.010)	0.168	0.653	0.980 (0.014)	0.970 (0.017)
lrcBART: $w = 0.9$, 100	0.065 (0.017)	0.183 (0.010)	1.262	5.920	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.065 (0.018)	0.189 (0.010)	1.268	5.950	1.000 (0.000)	0.000 (0.000)
BART-CP	0.056 (0.017)	0.176 (0.010)	0.168	0.657	0.930 (0.026)	0.950 (0.022)
LM-CP	0.290 (0.027)	0.395 (0.024)	0.273	1.067	0.850 (0.036)	0.910 (0.029)
hier. LM	0.357 (0.027)	0.444 (0.024)	0.588	2.243	1.000 (0.000)	0.500 (0.050)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $\text{IG}(3/2, 3s_\tau^2/2)$. The single-arm hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.05$.

E.9 Survival single-arm results

Both trial sizes and both scenarios are reported, first for median-survival ratios and then for five-year RMST ratios with floored control denominators (Appendix D.5).

Table S8: Complete Survival single arm simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: $n_{\text{trt}} = 30$, Sc1</i>						
lrcBART: 100	0.007 (0.044)	0.440 (0.046)	0.877	3.177	0.980 (0.014)	0.050 (0.022)
lrcBART: f90	0.011 (0.044)	0.439 (0.047)	0.906	3.271	1.000 (0.000)	0.050 (0.022)
lrcBART: s0min	0.007 (0.044)	0.440 (0.046)	0.877	3.177	0.980 (0.014)	0.050 (0.022)
lrcBART: $w = 0.9$, 100	0.016 (0.044)	0.442 (0.047)	3.323	10.974	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.016 (0.045)	0.443 (0.049)	3.220	10.974	1.000 (0.000)	0.000 (0.000)
BART-CP	0.044 (0.043)	0.430 (0.045)	0.894	3.292	0.990 (0.010)	0.060 (0.024)
AFT-CP	0.127 (0.065)	0.662 (0.074)	0.876	3.282	0.950 (0.022)	0.150 (0.036)
hier. AFT	0.177 (0.068)	0.697 (0.083)	1.862	5.683	1.000 (0.000)	0.040 (0.020)
<i>Median-survival ratio: $n_{\text{trt}} = 30$, Sc2</i>						
lrcBART: 100	0.023 (0.050)	0.496 (0.064)	1.008	3.672	0.990 (0.010)	0.070 (0.026)
lrcBART: f90	0.026 (0.050)	0.497 (0.067)	1.047	3.763	0.990 (0.010)	0.050 (0.022)
lrcBART: s0min	0.023 (0.050)	0.496 (0.064)	1.008	3.672	0.990 (0.010)	0.070 (0.026)
lrcBART: $w = 0.9$, 100	0.019 (0.050)	0.496 (0.064)	4.526	12.499	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.032 (0.050)	0.500 (0.067)	4.571	12.498	1.000 (0.000)	0.000 (0.000)
BART-CP	0.046 (0.048)	0.484 (0.057)	1.041	3.720	1.000 (0.000)	0.060 (0.024)
AFT-CP	0.326 (0.081)	0.867 (0.186)	1.053	3.918	0.980 (0.014)	0.210 (0.041)
hier. AFT	0.530 (0.092)	1.060 (0.218)	2.632	7.275	0.990 (0.010)	0.090 (0.029)
<i>Median-survival ratio: $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	-0.020 (0.018)	0.178 (0.013)	0.240	0.934	0.990 (0.010)	0.590 (0.049)
lrcBART: f90	-0.022 (0.018)	0.183 (0.013)	0.229	0.892	0.990 (0.010)	0.650 (0.048)
lrcBART: s0min	-0.021 (0.018)	0.178 (0.013)	0.215	0.838	0.970 (0.017)	0.680 (0.047)
lrcBART: $w = 0.9$, 100	-0.022 (0.018)	0.178 (0.013)	2.431	10.097	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	-0.021 (0.018)	0.183 (0.013)	2.420	10.166	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.011 (0.018)	0.180 (0.013)	0.217	0.849	0.970 (0.017)	0.730 (0.044)
AFT-CP	0.055 (0.037)	0.373 (0.027)	0.412	1.603	0.970 (0.017)	0.370 (0.048)
hier. AFT	0.097 (0.038)	0.388 (0.028)	1.386	4.191	1.000 (0.000)	0.020 (0.014)

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Table S8 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	0.070 (0.021)	0.216 (0.017)	0.257	1.003	0.980 (0.014)	0.730 (0.044)
lrcBART: f90	0.079 (0.021)	0.227 (0.016)	0.278	1.087	0.990 (0.010)	0.740 (0.044)
lrcBART: s0min	0.070 (0.021)	0.216 (0.017)	0.255	0.994	0.980 (0.014)	0.740 (0.044)
lrcBART: $w = 0.9$, 100	0.071 (0.020)	0.216 (0.017)	3.026	11.630	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.079 (0.021)	0.227 (0.016)	3.040	11.782	1.000 (0.000)	0.000 (0.000)
BART-CP	0.069 (0.021)	0.216 (0.015)	0.253	0.986	0.990 (0.010)	0.770 (0.042)
AFT-CP	0.676 (0.053)	0.859 (0.058)	0.668	2.583	0.820 (0.038)	0.810 (0.039)
hier. AFT	0.872 (0.057)	1.042 (0.061)	2.117	6.476	1.000 (0.000)	0.370 (0.048)
<i>Modified RMST ratio: $n_{\text{trt}} = 30$, Sc1</i>						
lrcBART: 100	0.007 (0.016)	0.161 (0.011)	0.211	0.825	0.990 (0.010)	0.140 (0.035)
lrcBART: f90	0.006 (0.016)	0.161 (0.011)	0.219	0.851	0.990 (0.010)	0.110 (0.031)
lrcBART: s0min	0.007 (0.016)	0.161 (0.011)	0.211	0.825	0.990 (0.010)	0.140 (0.035)
lrcBART: $w = 0.9$, 100	0.007 (0.016)	0.160 (0.011)	1.023	2.748	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.004 (0.016)	0.160 (0.011)	1.039	2.810	1.000 (0.000)	0.000 (0.000)
BART-CP	0.011 (0.016)	0.161 (0.011)	0.213	0.830	0.980 (0.014)	0.140 (0.035)
AFT-CP	-0.034 (0.018)	0.178 (0.013)	0.175	0.684	0.960 (0.020)	0.130 (0.034)
hier. AFT	-0.024 (0.018)	0.178 (0.013)	0.367	1.079	0.990 (0.010)	0.040 (0.020)
<i>Modified RMST ratio: $n_{\text{trt}} = 30$, Sc2</i>						
lrcBART: 100	-0.004 (0.012)	0.117 (0.008)	0.166	0.646	1.000 (0.000)	0.060 (0.024)
lrcBART: f90	-0.004 (0.012)	0.116 (0.009)	0.171	0.667	0.990 (0.010)	0.060 (0.024)
lrcBART: s0min	-0.004 (0.012)	0.117 (0.008)	0.166	0.646	1.000 (0.000)	0.060 (0.024)
lrcBART: $w = 0.9$, 100	-0.006 (0.012)	0.116 (0.008)	0.755	1.985	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	-0.004 (0.012)	0.115 (0.008)	0.779	2.023	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.006 (0.011)	0.114 (0.009)	0.164	0.641	1.000 (0.000)	0.050 (0.022)
AFT-CP	0.050 (0.014)	0.148 (0.013)	0.160	0.627	0.970 (0.017)	0.220 (0.041)
hier. AFT	0.085 (0.014)	0.166 (0.015)	0.486	1.057	0.990 (0.010)	0.100 (0.030)
<i>Modified RMST ratio: $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	-0.008 (0.006)	0.061 (0.004)	0.083	0.323	0.990 (0.010)	0.490 (0.050)
lrcBART: f90	-0.008 (0.006)	0.064 (0.005)	0.079	0.309	0.980 (0.014)	0.530 (0.050)
lrcBART: s0min	-0.008 (0.006)	0.061 (0.004)	0.075	0.295	0.980 (0.014)	0.570 (0.050)
lrcBART: $w = 0.9$, 100	-0.009 (0.006)	0.062 (0.005)	0.816	2.370	1.000 (0.000)	0.000 (0.000)

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Table S8 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: $w = 0.9$, f90	-0.009 (0.006)	0.064 (0.005)	0.815	2.370	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.005 (0.006)	0.062 (0.005)	0.075	0.295	0.980 (0.014)	0.550 (0.050)
AFT-CP	-0.013 (0.010)	0.096 (0.006)	0.105	0.410	0.970 (0.017)	0.330 (0.047)
hier. AFT	-0.004 (0.010)	0.096 (0.006)	0.411	0.944	1.000 (0.000)	0.020 (0.014)
<i>Modified RMST ratio: $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	0.029 (0.007)	0.074 (0.005)	0.079	0.307	0.960 (0.020)	0.650 (0.048)
lrcBART: f90	0.032 (0.007)	0.076 (0.005)	0.084	0.328	0.960 (0.020)	0.620 (0.049)
lrcBART: s0min	0.029 (0.007)	0.074 (0.005)	0.078	0.305	0.960 (0.020)	0.660 (0.047)
lrcBART: $w = 0.9$, 100	0.029 (0.007)	0.073 (0.005)	0.853	2.334	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.032 (0.007)	0.076 (0.005)	0.847	2.353	1.000 (0.000)	0.000 (0.000)
BART-CP	0.029 (0.007)	0.075 (0.005)	0.077	0.303	0.950 (0.022)	0.700 (0.046)
AFT-CP	0.155 (0.010)	0.186 (0.010)	0.129	0.503	0.810 (0.039)	0.830 (0.038)
hier. AFT	0.185 (0.010)	0.211 (0.010)	0.465	1.122	1.000 (0.000)	0.370 (0.048)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $\text{IG}(3/2, 3s_\tau^2/2)$. The single-arm hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.05$.

E.10 Prior ESS sensitivity and regional summaries

Table S8 distinguishes the requested target, the target after ceiling capping, the block-based curve value and the selected prior scale. These summarize per-dataset prior ESS calibration. The regional-discrepancy figure summarizes the posterior probability of a small discrepancy and posterior mean discrepancy are averaged over trial-control profiles in half-unit bins on X_5 . It uses the first 40 Gaussian replicates for three regional shifts of 0.5, 1 and 2 (Sc5). For each bin, profile-specific posterior summaries are pooled across replicates and averaged; replicates contribute in proportion to their trial-control counts within that bin. The figure shows binned means, not credible intervals or replicate-variability intervals.

Table S9: Replicate-averaged lrcBART prior ESS calibration summaries.

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
<i>gaussian, Sc1</i>						
lrcBART: 100	100.0	96.4	82.5	134.5	0.21	0.001626
lrcBART: 75	75.0	74.4	62.4	134.5	0.05	0.003083
lrcBART: 50	50.0	50.0	42.6	134.5	0.00	0.006187
lrcBART: f90	121.1	121.1	96.0	134.5	0.00	0.000967
lrcBART: f50	67.3	67.3	56.6	134.5	0.00	0.003845
lrcBART: f25	33.6	33.6	29.6	134.5	0.00	0.011093
<i>gaussian, Sc2</i>						
lrcBART: 100	100.0	22.8	24.2	22.8	1.00	0.000001
lrcBART: 75	75.0	22.8	24.2	22.8	1.00	0.000001
lrcBART: 50	50.0	22.8	24.0	22.8	0.99	0.000036
lrcBART: f90	20.5	20.5	16.0	22.8	0.00	0.005771
lrcBART: f50	11.4	11.4	9.5	22.8	0.00	0.021554
lrcBART: f25	5.7	5.7	5.0	22.8	0.00	0.060643
<i>gaussian, Sc3, $\rho = -0.5$</i>						
lrcBART: 100	100.0	99.3	82.6	150.1	0.06	0.003895
lrcBART: 75	75.0	75.0	63.1	150.1	0.00	0.006945
lrcBART: 50	50.0	50.0	43.3	150.1	0.00	0.013433
lrcBART: f90	135.1	135.1	108.1	150.1	0.00	0.001726
lrcBART: f50	75.0	75.0	63.3	150.1	0.00	0.007055
lrcBART: f25	37.5	37.5	33.1	150.1	0.00	0.020388
<i>gaussian, Sc3, $\rho = 0$</i>						
lrcBART: 100	100.0	100.0	84.4	220.1	0.00	0.005907
lrcBART: 75	75.0	75.0	64.6	220.1	0.00	0.009315
lrcBART: 50	50.0	50.0	44.0	220.1	0.00	0.016348
lrcBART: f90	198.1	198.1	156.5	220.1	0.00	0.001284
lrcBART: f50	110.1	110.1	91.8	220.1	0.00	0.005124
lrcBART: f25	55.0	55.0	48.6	220.1	0.00	0.014482
<i>gaussian, Sc3, $\rho = 0.5$</i>						
lrcBART: 100	100.0	99.3	82.6	148.4	0.06	0.003844
lrcBART: 75	75.0	75.0	63.0	148.4	0.00	0.006947
lrcBART: 50	50.0	50.0	43.4	148.4	0.00	0.013417
lrcBART: f90	133.6	133.6	106.8	148.4	0.00	0.001772
lrcBART: f50	74.2	74.2	62.5	148.4	0.00	0.007267
lrcBART: f25	37.1	37.1	32.8	148.4	0.00	0.020901

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
<i>gaussian, Sc4, $\delta = 1$</i>						
lrcBART: 100	100.0	96.4	82.5	134.5	0.21	0.001626
lrcBART: 75	75.0	74.4	62.4	134.5	0.05	0.003083
lrcBART: 50	50.0	50.0	42.6	134.5	0.00	0.006187
lrcBART: f90	121.1	121.1	96.0	134.5	0.00	0.000967
lrcBART: f50	67.3	67.3	56.6	134.5	0.00	0.003845
lrcBART: f25	33.6	33.6	29.6	134.5	0.00	0.011093
<i>gaussian, Sc4, $\delta = 2$</i>						
lrcBART: 100	100.0	96.4	82.5	134.5	0.21	0.001626
lrcBART: 75	75.0	74.4	62.4	134.5	0.05	0.003083
lrcBART: 50	50.0	50.0	42.6	134.5	0.00	0.006187
lrcBART: f90	121.1	121.1	96.0	134.5	0.00	0.000967
lrcBART: f50	67.3	67.3	56.6	134.5	0.00	0.003845
lrcBART: f25	33.6	33.6	29.6	134.5	0.00	0.011093
<i>gaussian, Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	100.0	96.2	83.0	132.6	0.24	0.001592
lrcBART: 75	75.0	74.4	62.0	132.6	0.04	0.003091
lrcBART: 50	50.0	50.0	42.6	132.6	0.02	0.006189
lrcBART: f90	119.3	119.3	94.3	132.6	0.00	0.001020
lrcBART: f50	66.3	66.3	55.5	132.6	0.00	0.003963
lrcBART: f25	33.1	33.1	29.1	132.6	0.00	0.011303
<i>gaussian, Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	100.0	96.0	83.0	131.7	0.26	0.001577
lrcBART: 75	75.0	74.4	62.4	131.7	0.06	0.003036
lrcBART: 50	50.0	50.0	42.4	131.7	0.00	0.006162
lrcBART: f90	118.6	118.6	93.7	131.7	0.00	0.001014
lrcBART: f50	65.9	65.9	55.1	131.7	0.00	0.003970
lrcBART: f25	32.9	32.9	28.8	131.7	0.00	0.011422
<i>gaussian, Sc5, X_5, $\delta = 2$</i>						
lrcBART: 100	100.0	95.6	81.7	130.1	0.22	0.001570
lrcBART: 75	75.0	74.3	62.0	130.1	0.05	0.003030
lrcBART: 50	50.0	49.9	42.7	130.1	0.01	0.006075
lrcBART: f90	117.1	117.1	92.8	130.1	0.00	0.001005
lrcBART: f50	65.0	65.0	54.4	130.1	0.00	0.004028
lrcBART: f25	32.5	32.5	28.7	130.1	0.00	0.011446
<i>gaussian, Sc5, X_7, $\delta = 1$</i>						
lrcBART: 100	100.0	95.3	82.7	130.6	0.25	0.001542
lrcBART: 75	75.0	74.1	62.3	130.6	0.08	0.003018
lrcBART: 50	50.0	50.0	42.6	130.6	0.01	0.006072
lrcBART: f90	117.6	117.6	93.1	130.6	0.00	0.001030
lrcBART: f50	65.3	65.3	54.7	130.6	0.00	0.004042
lrcBART: f25	32.7	32.7	28.8	130.6	0.00	0.011468
<i>gaussian, Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	100.0	95.4	82.4	129.9	0.24	0.001544
lrcBART: 75	75.0	74.2	62.2	129.9	0.06	0.002990
lrcBART: 50	50.0	50.0	42.6	129.9	0.02	0.006085

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
lrcBART: f90	117.0	117.0	92.5	129.9	0.00	0.001021
lrcBART: f50	65.0	65.0	54.4	129.9	0.00	0.004030
lrcBART: f25	32.5	32.5	28.5	129.9	0.00	0.011577
<i>survival, Sc1</i>						
lrcBART: 100	100.0	99.3	83.2	202.3	0.05	0.004399
lrcBART: 75	75.0	74.9	63.7	202.3	0.01	0.007015
lrcBART: 50	50.0	50.0	43.6	202.3	0.00	0.012328
lrcBART: f90	182.1	182.1	142.5	202.3	0.00	0.001139
lrcBART: f50	101.2	101.2	84.0	202.3	0.00	0.004398
lrcBART: f25	50.6	50.6	44.3	202.3	0.00	0.012413
<i>survival, Sc2</i>						
lrcBART: 100	100.0	36.2	38.3	36.2	1.00	0.000001
lrcBART: 75	75.0	36.2	38.1	36.2	0.99	0.000029
lrcBART: 50	50.0	35.2	35.3	36.2	0.87	0.000756
lrcBART: f90	32.6	32.6	25.5	36.2	0.00	0.005652
lrcBART: f50	18.1	18.1	15.0	36.2	0.00	0.021673
lrcBART: f25	9.1	9.1	7.9	36.2	0.00	0.061216
<i>survival, Sc3, $\rho = -0.5$</i>						
lrcBART: 100	100.0	99.9	82.9	193.6	0.01	0.006562
lrcBART: 75	75.0	75.0	63.7	193.6	0.00	0.010583
lrcBART: 50	50.0	50.0	43.5	193.6	0.00	0.019003
lrcBART: f90	174.3	174.3	137.1	193.6	0.00	0.001825
lrcBART: f50	96.8	96.8	80.8	193.6	0.00	0.007121
lrcBART: f25	48.4	48.4	42.4	193.6	0.00	0.020292
<i>survival, Sc3, $\rho = 0$</i>						
lrcBART: 100	100.0	100.0	86.2	302.7	0.00	0.009624
lrcBART: 75	75.0	75.0	66.1	302.7	0.00	0.014262
lrcBART: 50	50.0	50.0	44.7	302.7	0.00	0.024000
lrcBART: f90	272.4	272.4	217.8	302.7	0.00	0.001224
lrcBART: f50	151.3	151.3	127.5	302.7	0.00	0.004992
lrcBART: f25	75.7	75.7	66.7	302.7	0.00	0.014404
<i>survival, Sc3, $\rho = 0.5$</i>						
lrcBART: 100	100.0	99.8	84.5	201.3	0.01	0.006783
lrcBART: 75	75.0	75.0	64.4	201.3	0.00	0.011035
lrcBART: 50	50.0	50.0	44.1	201.3	0.00	0.019587
lrcBART: f90	181.2	181.2	144.8	201.3	0.00	0.001706
lrcBART: f50	100.7	100.7	84.9	201.3	0.00	0.006934
lrcBART: f25	50.3	50.3	44.3	201.3	0.00	0.020040
<i>survival, Sc4, $\delta = 1$</i>						
lrcBART: 100	100.0	98.4	82.1	177.0	0.09	0.003993
lrcBART: 75	75.0	74.8	63.0	177.0	0.03	0.006510
lrcBART: 50	50.0	50.0	42.8	177.0	0.00	0.011885
lrcBART: f90	159.3	159.3	123.8	177.0	0.00	0.001388
lrcBART: f50	88.5	88.5	73.3	177.0	0.00	0.005147
lrcBART: f25	44.2	44.2	38.6	177.0	0.00	0.014486

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
<i>survival, Sc4, $\delta = 2$</i>						
lrcBART: 100	100.0	95.1	81.2	141.1	0.25	0.003228
lrcBART: 75	75.0	74.1	61.0	141.1	0.06	0.005752
lrcBART: 50	50.0	50.0	42.0	141.1	0.01	0.010976
lrcBART: f90	127.0	127.0	97.4	141.1	0.00	0.001905
lrcBART: f50	70.5	70.5	57.8	141.1	0.00	0.006715
lrcBART: f25	35.3	35.3	30.7	141.1	0.00	0.018381
<i>survival, Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	100.0	98.9	82.7	191.6	0.04	0.004175
lrcBART: 75	75.0	74.8	63.4	191.6	0.02	0.006801
lrcBART: 50	50.0	50.0	43.4	191.6	0.00	0.012144
lrcBART: f90	172.4	172.4	136.3	191.6	0.00	0.001182
lrcBART: f50	95.8	95.8	80.1	191.6	0.00	0.004644
lrcBART: f25	47.9	47.9	42.1	191.6	0.00	0.013207
<i>survival, Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	100.0	99.0	82.8	181.8	0.08	0.004062
lrcBART: 75	75.0	74.8	63.1	181.8	0.02	0.006621
lrcBART: 50	50.0	50.0	43.2	181.8	0.00	0.011961
lrcBART: f90	163.6	163.6	128.1	181.8	0.00	0.001291
lrcBART: f50	90.9	90.9	75.6	181.8	0.00	0.004925
lrcBART: f25	45.5	45.5	39.8	181.8	0.00	0.013909
<i>survival, Sc5, X_5, $\delta = 2$</i>						
lrcBART: 100	100.0	97.5	82.4	159.4	0.18	0.003622
lrcBART: 75	75.0	74.5	61.8	159.4	0.03	0.006196
lrcBART: 50	50.0	50.0	42.6	159.4	0.01	0.011458
lrcBART: f90	143.5	143.5	110.5	159.4	0.00	0.001598
lrcBART: f50	79.7	79.7	65.5	159.4	0.00	0.005816
lrcBART: f25	39.9	39.9	34.6	159.4	0.00	0.016210
<i>survival, Sc5, X_7, $\delta = 1$</i>						
lrcBART: 100	100.0	98.9	82.8	186.2	0.07	0.004149
lrcBART: 75	75.0	74.8	63.3	186.2	0.02	0.006717
lrcBART: 50	50.0	50.0	43.3	186.2	0.00	0.012069
lrcBART: f90	167.6	167.6	131.6	186.2	0.00	0.001245
lrcBART: f50	93.1	93.1	77.7	186.2	0.00	0.004752
lrcBART: f25	46.6	46.6	40.9	186.2	0.00	0.013513
<i>survival, Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	100.0	98.3	82.5	167.2	0.12	0.003742
lrcBART: 75	75.0	74.7	62.1	167.2	0.02	0.006364
lrcBART: 50	50.0	50.0	42.9	167.2	0.00	0.011595
lrcBART: f90	150.5	150.5	117.1	167.2	0.00	0.001461
lrcBART: f50	83.6	83.6	69.2	167.2	0.00	0.005431
lrcBART: f25	41.8	41.8	36.5	167.2	0.00	0.015186
<i>gaussian single arm, $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	100.0	100.0	81.4	152.8	0.00	0.001838
lrcBART: f90	137.6	137.6	108.5	152.8	0.00	0.000815
lrcBART: s0min	–	–	160.6	152.8	0.00	0.000001

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
lrcBART: $w = 0.9$, 100	100.0	100.0	81.4	152.8	0.00	0.001838
lrcBART: $w = 0.9$, f90	137.6	137.6	108.5	152.8	0.00	0.000815
<i>gaussian single arm, $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	100.0	97.9	84.2	109.0	0.29	0.000891
lrcBART: f90	98.1	98.1	77.0	109.0	0.00	0.001173
lrcBART: s0min	–	–	114.6	109.0	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	97.9	84.2	109.0	0.29	0.000891
lrcBART: $w = 0.9$, f90	98.1	98.1	77.0	109.0	0.00	0.001173
<i>survival single arm, $n_{\text{trt}} = 30$, Sc1</i>						
lrcBART: 100	100.0	55.9	60.5	55.9	1.00	0.000001
lrcBART: f90	50.3	50.3	38.3	55.9	0.00	0.002037
lrcBART: s0min	–	–	60.5	55.9	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	55.9	60.5	55.9	1.00	0.000001
lrcBART: $w = 0.9$, f90	50.3	50.3	38.3	55.9	0.00	0.002037
<i>survival single arm, $n_{\text{trt}} = 30$, Sc2</i>						
lrcBART: 100	100.0	41.9	45.6	41.9	1.00	0.000001
lrcBART: f90	37.7	37.7	28.6	41.9	0.00	0.002817
lrcBART: s0min	–	–	45.6	41.9	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	41.9	45.6	41.9	1.00	0.000001
lrcBART: $w = 0.9$, f90	37.7	37.7	28.6	41.9	0.00	0.002817
<i>survival single arm, $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	100.0	99.9	79.9	133.4	0.03	0.001190
lrcBART: f90	120.0	120.0	93.1	133.4	0.00	0.000755
lrcBART: s0min	–	–	141.8	133.4	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	99.9	79.9	133.4	0.03	0.001190
lrcBART: $w = 0.9$, f90	120.0	120.0	93.1	133.4	0.00	0.000755
<i>survival single arm, $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	100.0	83.5	85.7	84.7	0.88	0.000099
lrcBART: f90	76.2	76.2	59.3	84.7	0.00	0.001206
lrcBART: s0min	–	–	90.2	84.7	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	83.5	85.7	84.7	0.88	0.000099
lrcBART: $w = 0.9$, f90	76.2	76.2	59.3	84.7	0.00	0.001206

All quantities are means over 100 datasets, except cap fraction, which is the proportion flagged as capped. Curve ESS is the mean across block-specific ESS calculations at the selected scale. The ceiling uses the variance across all historical posterior draws. These are numerical summaries of distinct rules (Appendix B.5); neither is a posterior borrowing count. s0min bypasses target selection. A dash denotes an unavailable or inapplicable quantity.

F Application details and supplementary results

F.1 Analysis extract and harmonized variables

The available merged analysis extract has 30 EloKRd and 200 triplet-treated UCMM observations. Starting from 797 UCMM records, the preparation code excludes negative OS/PFS times (790 remain), requires a recorded induction start (303), excludes E-Rd/Elo-Rd (302), and retains verified three-agent regimens (200). The EloKRd analysis uses the available subset of the published study (Derman et al., 2022).

Triplet-only C3 selection. Figure S2 shows the sequential exclusions. C2 contains 302 patients after excluding E-Rd/Elo-Rd. The triplet rule then excludes 102 patients with doublets ($n = 66$), single-agent treatment ($n = 4$), four or more treatment agents ($n = 19$), or other/unclear treatment records ($n = 13$). C3 therefore contains 200 UCMM controls, or 230 observations when merged with the 30 EloKRd patients. Classification uses the recorded induction components rather than the source triplet flag. Steroids count as treatment agents; zoledronic acid is supportive therapy. VRd plus vorinostat is excluded because it contains four treatment agents, despite a source triplet flag.

The retained regimen labels and counts are VRd (107), KRd (38), CyBorD (22), CRd (5), Cy-Kd (4), IRd (4), KPd (4), VTD (4), PDd (3), VPd (2), and one each of BRd, Dara-Rd, Elo-Vd, O-Rd, VAD, VMP and VRd-Zoledronic. In this extract, PDd records bortezomib, doxorubicin and dexamethasone; O-Rd records oprozomib, lenalidomide and dexamethasone. The 13 other/unclear records are not assumed to be triplets. Covariate harmonization retains all 200 analysis controls.

The EloKRd extract supplies times from first treatment (C1D1); the UCMM code constructs times from induction start. OS ends at death or last contact. UCMM PFS ends at the first progression or death, using the recorded first-progression date with a recurrence-date fallback; otherwise it ends at last contact. Zero UCMM PFS times are floored at 0.1 month before logarithmic analysis. EloKRd PFS status 2 (death without prior progression) is recoded as an event. Differences in source-specific endpoint ascertainment may remain despite this harmonization.

Variable selection first retains variables complete in EloKRd, then drops variables with the greatest UCMM missingness until all 200 triplet-treated UCMM records are complete on the retained set. The final numeric model predictors have no missing values. This data-dependent availability rule should be distinguished from a clinically prespecified adjustment set.

F.2 Analysis and sensitivity configurations

All adjusted methods use the same harmonized predictors and standardize control and treated predictions to the 30 EloKRd profiles. KM remains unadjusted. Table S10 reports the hypothetical EloKRd control for adjusted methods and the observed UCMM cohort RMST for KM. Ratios and differences use those respective controls. The primary lrcBART analysis uses $H_f = 10$, $H_g = 5$, fixed $w = 1$ and requested ESS 30. Sensitivities cross $H_f \in \{10, 50\}$ with fixed $w \in \{1, 0.9\}$. Each of those four settings uses six requested targets: 30, 22.5, 15 and fractions 0.90, 0.50, 0.25 of its calibration ceiling. The target labels 100, 75 and 50 correspond to percentages of the 30-patient trial here; in simulations those labels are absolute counts.

The five comparator settings are KM, AFT-CP, BART-CP and hierarchical AFT at discrepancy-variance prior scales $s_\tau^2 = 0.05$ and $s_\tau^2 = 0.5$. Together with 24 lrcBART settings this gives 29 rows per endpoint.

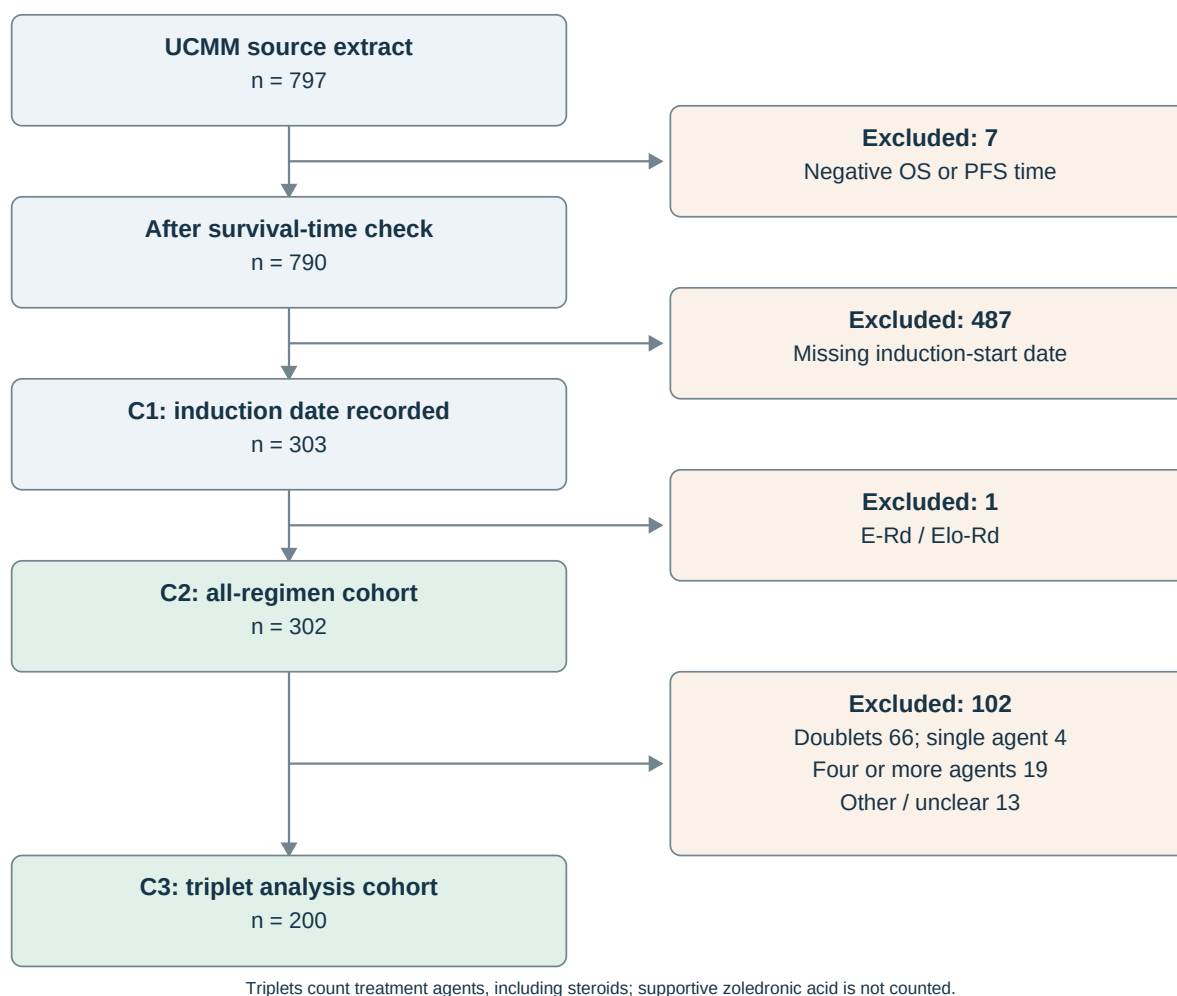


Figure S2: UCMM inclusion and exclusion flow through C2 and the triplet-only analysis cohort. Counts refer to sequential, mutually exclusive exclusions.

Table S10: Harmonized model variables in the myeloma application.

Dimension	Numeric model representation and coding
Age	Continuous age in years: enrollment age for EloKRd; age at the identification date for UCMM, with diagnosis-age fallback. Table 7 groups rows into < 50 , $[50, 60)$, $[60, 70)$ and ≥ 70 years for display, shows age to one decimal year, and uses exact age in every calculation.
Sex	Male indicator: 1 for male, 0 for female; source strings beginning with M or F are harmonized.
Race	Black and Other indicators, with White as reference. White/Caucasian and Black/African labels are combined. Other comprises remaining harmonized categories, including Other/Unknown; it is not an imputation of a known race.
Hispanic/Latino	1 for “Hispanic or Latino”; 0 for the non-Hispanic labels in the extract. Ethnicity remains a separate dimension from race.
Cytogenetic risk	EloKRd uses its recorded 0/1 high-risk flag. UCMM uses HRCA ≥ 1 when recorded; otherwise any positive t(4;14), t(14;16), t(14;20), del(17p) or gain(1q) indicator gives 1. In that fallback, missing component indicators are ignored and no positive entry gives 0.
ASCT	1 for transplantation, 0 otherwise, using the cleaned transplantation indicators. In EloKRd the source variable records off-protocol transplantation. This is a postinduction characteristic.
Outcome records	PFS and OS times converted from months to years; event indicator 1 and right-censoring indicator 0.

Seven numeric predictors represent six covariate dimensions. The final extract has no missing numeric model predictors. The harmonized variables can retain source-specific measurement differences.

The main table reports $s^2_{\tau} = 0.05$; both hierarchical settings remain in the complete appendix table. RMST contrasts use the draw-by-draw calculations in Appendix D, with posterior medians and equal-tailed 95% intervals.

F.3 Complete application results

All 58 PFS and OS results are included. Similarity among lrcBART estimates does not resolve prior dependence for unobserved trial-control outcomes.

Table S11: Complete application results, including all lrcBART sensitivity settings.

Method	Setting	Ratio (95% interval)	Difference (95% interval)	EloKRd treatment RMST	EloKRd hypothetical control RMST	UCMM control RMST
<i>PFS</i>						
KM	—	1.040 (0.862, 1.254)	0.143 (-0.548, 0.834)	3.705	—	3.562
lrcBART	default, 100	0.997 (0.799, 1.229)	-0.009 (-0.768, 0.714)	3.521	3.535	—
lrcBART	default, 75	0.998 (0.792, 1.250)	-0.009 (-0.806, 0.760)	3.521	3.534	—
lrcBART	default, 50	0.997 (0.779, 1.311)	-0.011 (-0.883, 0.893)	3.521	3.536	—
lrcBART	default, f90	0.997 (0.801, 1.224)	-0.012 (-0.759, 0.702)	3.521	3.534	—
lrcBART	default, f50	0.998 (0.786, 1.280)	-0.008 (-0.847, 0.820)	3.521	3.534	—
lrcBART	default, f25	0.997 (0.760, 1.419)	-0.011 (-1.000, 1.085)	3.521	3.529	—
lrcBART	Hf50, 100	0.987 (0.798, 1.220)	-0.046 (-0.775, 0.684)	3.521	3.572	—
lrcBART	Hf50, 75	0.987 (0.792, 1.247)	-0.049 (-0.804, 0.752)	3.521	3.572	—
lrcBART	Hf50, 50	0.988 (0.782, 1.290)	-0.044 (-0.875, 0.849)	3.521	3.571	—
lrcBART	Hf50, f90	0.987 (0.798, 1.222)	-0.046 (-0.775, 0.689)	3.521	3.572	—
lrcBART	Hf50, f50	0.987 (0.783, 1.281)	-0.047 (-0.865, 0.829)	3.521	3.570	—
lrcBART	Hf50, f25	0.991 (0.754, 1.436)	-0.032 (-1.021, 1.131)	3.521	3.562	—
lrcBART	w0.9, 100	0.996 (0.738, 1.754)	-0.015 (-1.131, 1.526)	3.521	3.539	—
lrcBART	w0.9, 75	0.995 (0.736, 1.753)	-0.017 (-1.145, 1.543)	3.521	3.537	—
lrcBART	w0.9, 50	0.996 (0.730, 1.783)	-0.014 (-1.176, 1.578)	3.521	3.538	—
lrcBART	w0.9, f90	0.995 (0.740, 1.750)	-0.018 (-1.129, 1.535)	3.521	3.539	—
lrcBART	w0.9, f50	0.996 (0.733, 1.767)	-0.014 (-1.163, 1.552)	3.521	3.537	—
lrcBART	w0.9, f25	0.997 (0.725, 1.842)	-0.012 (-1.223, 1.655)	3.521	3.537	—
lrcBART	Hf50, w0.9, 100	0.984 (0.734, 1.735)	-0.057 (-1.161, 1.545)	3.521	3.582	—
lrcBART	Hf50, w0.9, 75	0.985 (0.731, 1.737)	-0.053 (-1.170, 1.545)	3.521	3.584	—
lrcBART	Hf50, w0.9, 50	0.986 (0.730, 1.765)	-0.053 (-1.179, 1.563)	3.521	3.581	—
lrcBART	Hf50, w0.9, f90	0.985 (0.734, 1.733)	-0.057 (-1.160, 1.543)	3.521	3.582	—
lrcBART	Hf50, w0.9, f50	0.985 (0.730, 1.765)	-0.054 (-1.180, 1.561)	3.521	3.583	—
lrcBART	Hf50, w0.9, f25	0.987 (0.720, 1.844)	-0.048 (-1.244, 1.673)	3.521	3.570	—
BART-CP	—	0.988 (0.796, 1.215)	-0.042 (-0.760, 0.690)	3.527	3.580	—
AFT-CP	—	0.901 (0.708, 1.105)	-0.338 (-1.080, 0.335)	3.069	3.408	—
hier. AFT	—	1.062 (0.789, 1.467)	0.192 (-0.834, 1.138)	3.355	3.148	—
hier. AFT	—	1.209 (0.731, 2.136)	0.567 (-1.204, 1.784)	3.355	2.715	—

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Table S11 (continued)

Method	Setting	Ratio (95% interval)	Difference (95% interval)	EloKRd treatment RMST	EloKRd hypothetical control RMST	UCMM control RMST
<i>OS</i>						
KM	—	0.891 (0.776, 1.023)	-0.505 (-1.077, 0.068)	4.122	—	4.627
lrcBART	default, 100	0.898 (0.765, 1.024)	-0.453 (-1.065, 0.101)	3.995	4.456	—
lrcBART	default, 75	0.898 (0.764, 1.034)	-0.454 (-1.073, 0.142)	3.995	4.453	—
lrcBART	default, 50	0.899 (0.763, 1.051)	-0.451 (-1.090, 0.207)	3.995	4.452	—
lrcBART	default, f90	0.898 (0.766, 1.030)	-0.454 (-1.068, 0.123)	3.995	4.455	—
lrcBART	default, f50	0.899 (0.763, 1.054)	-0.452 (-1.096, 0.216)	3.995	4.452	—
lrcBART	default, f25	0.900 (0.756, 1.120)	-0.444 (-1.149, 0.440)	3.995	4.446	—
lrcBART	Hf50, 100	0.901 (0.771, 1.024)	-0.439 (-1.039, 0.100)	3.995	4.434	—
lrcBART	Hf50, 75	0.902 (0.768, 1.037)	-0.437 (-1.052, 0.154)	3.995	4.432	—
lrcBART	Hf50, 50	0.902 (0.766, 1.057)	-0.432 (-1.064, 0.229)	3.995	4.428	—
lrcBART	Hf50, f90	0.901 (0.769, 1.032)	-0.436 (-1.043, 0.132)	3.995	4.432	—
lrcBART	Hf50, f50	0.902 (0.766, 1.061)	-0.431 (-1.068, 0.242)	3.995	4.427	—
lrcBART	Hf50, f25	0.906 (0.759, 1.141)	-0.415 (-1.121, 0.513)	3.995	4.417	—
lrcBART	w0.9, 100	0.897 (0.763, 1.067)	-0.458 (-1.098, 0.261)	3.995	4.457	—
lrcBART	w0.9, 75	0.897 (0.762, 1.072)	-0.458 (-1.110, 0.282)	3.995	4.457	—
lrcBART	w0.9, 50	0.897 (0.759, 1.087)	-0.456 (-1.122, 0.334)	3.995	4.455	—
lrcBART	w0.9, f90	0.897 (0.762, 1.067)	-0.459 (-1.104, 0.268)	3.995	4.458	—
lrcBART	w0.9, f50	0.897 (0.759, 1.090)	-0.457 (-1.123, 0.348)	3.995	4.454	—
lrcBART	w0.9, f25	0.899 (0.754, 1.156)	-0.454 (-1.154, 0.558)	3.995	4.452	—
lrcBART	Hf50, w0.9, 100	0.902 (0.766, 1.068)	-0.432 (-1.080, 0.265)	3.995	4.434	—
lrcBART	Hf50, w0.9, 75	0.903 (0.765, 1.082)	-0.432 (-1.072, 0.318)	3.995	4.432	—
lrcBART	Hf50, w0.9, 50	0.904 (0.765, 1.099)	-0.425 (-1.088, 0.371)	3.995	4.428	—
lrcBART	Hf50, w0.9, f90	0.903 (0.766, 1.076)	-0.432 (-1.067, 0.297)	3.995	4.433	—
lrcBART	Hf50, w0.9, f50	0.904 (0.765, 1.103)	-0.424 (-1.091, 0.389)	3.995	4.428	—
lrcBART	Hf50, w0.9, f25	0.907 (0.758, 1.174)	-0.413 (-1.136, 0.610)	3.995	4.418	—
BART-CP	—	0.898 (0.769, 1.017)	-0.448 (-1.048, 0.073)	3.985	4.439	—
AFT-CP	—	0.809 (0.654, 0.951)	-0.814 (-1.521, -0.203)	3.449	4.268	—
hier. AFT	—	0.961 (0.761, 1.387)	-0.153 (-1.085, 1.073)	3.807	3.953	—
hier. AFT	—	1.208 (0.777, 1.710)	0.646 (-1.057, 1.684)	3.807	3.117	—

All 58 method/endpoint results are included. Default: $H_f = 10$, $H_g = 5$, $w = 1$. Hf50 changes H_f to 50; w0.9 fixes $w = 0.9$. Application labels 100, 75 and 50 request ESS 30, 22.5 and 15, respectively; f90, f50 and f25 request the corresponding fraction of the calibration ceiling. Intervals, units and arm-column definitions follow Table 6. Contrasts use the EloKRd hypothetical control for adjusted methods and the observed UCMM control for KM. Within each endpoint, the first and second hier. AFT rows use hierarchical discrepancy-variance prior scales $s_\tau^2 = 0.05$ and $s_\tau^2 = 0.5$, respectively.

F.4 Construction of the patient-profile PFS table

Table 7 contains one row for each of the 30 EloKRd patients. Rows are ordered by age group (< 50 , $50\text{--}59$, $60\text{--}69$ and ≥ 70 years) and exact age. The age group and UCMM count are displayed separately on every row. A row retains that patient's complete model profile: exact age, sex, race, Hispanic/Latino status, cytogenetic risk and ASCT. Age is rounded to one decimal year in the display, while all calculations use its exact value. The other predictors are not set to common reference values.

The preliminary historical model is the primary PFS specification with $H_f = 10$. Its 4,000 retained posterior predictions at the EloKRd profiles give $V_f(x_i)$ directly as the sample variance across draws for each patient. For a single profile, one leaf from each discrepancy tree has weight one, so the all-spike discrepancy variance is $H_g \tau_0^2$. The expectation uses 100,000 chi-square draws with $\nu_0 = 3$ and the globally selected $s_0^2 = 0.00398107$.

For each row, the UCMM cohort count is obtained by matching historical patients on age group, sex, race, Hispanic/Latino status, cytogenetic risk and ASCT. Counts repeat when EloKRd rows share a displayed combination and should not be summed down the table. A count is descriptive and does not restrict the historical fit used for that row's ESS. Every profile uses the same historical residual-variance reference and scale law. Table 7 evaluates the full-variance expectation-of-ratios definition using the preliminary historical fit. ESS is reported to one decimal place, ranges from 3.8 to 24.5, and is expressed in uncensored normal-reference observations for the conditional mean log-PFS at the same profile. Only PFS appears in this table; OS remains in the outcome tables.

F.5 Prior ESS calibration and MCMC diagnostics

The preliminary application calibration uses 2,000 warm-up and 4,000 retained draws, 2,000 prior trees, 4,000 scale draws and the 141-value scale grid. Table S11(a) reports requested/capped targets, selected scales, the block-based curve and the full-variance ESS evaluated with the estimated prior-tree weight expectation. The latter uses 100,000 deterministic chi-square quantiles for numerical integration. Differences reflect block averaging and numerical integration error; the reciprocal-variance ordering in Appendix B applies with common scale draws.

Each lrcBART outcome fit uses four chains with 2,000 warm-up and 2,000 retained draws per chain. Table S11(b) reports $\hat{R}$, bulk ESS and tail ESS for the log RMST ratio. Prior ESS and MCMC ESS have different meanings: the former concerns equivalent reference information about a specified current-control mean, whereas the latter concerns simulation efficiency. These scalar diagnostics do not establish exploration of all tree structures.

Table S11(a): Application prior ESS calibration.

Target label	Requested	Capped target	s_0^2	Block-based ESS	Full-variance ESS	Ceiling
<i>PFS, $H_f = 10$</i>						
100	30.00	30.00	0.003981	24.70	24.43	37.03
75	22.50	22.50	0.007943	19.40	19.28	37.03
50	15.00	15.00	0.017783	13.07	13.08	37.03
f90	33.33	33.33	0.002818	27.07	26.71	37.03
f50	18.51	18.51	0.012589	15.72	15.69	37.03
f25	9.26	9.26	0.035481	8.43	8.47	37.03
<i>PFS, $H_f = 50$</i>						
100	30.00	30.00	0.001585	27.22	27.09	32.97
75	22.50	22.50	0.005623	20.28	20.25	32.97
50	15.00	15.00	0.014125	13.90	13.94	32.97
f90	29.67	29.67	0.001778	26.73	26.60	32.97
f50	16.48	16.48	0.012589	14.70	14.73	32.97
f25	8.24	8.24	0.039811	7.47	7.52	32.97
<i>OS, $H_f = 10$</i>						
100	30.00	29.14	0.000001	30.04	29.13	29.14
75	22.50	22.50	0.007079	18.93	18.66	29.14
50	15.00	15.00	0.017783	13.21	13.13	29.14
f90	26.22	26.22	0.003981	22.06	21.63	29.14
f50	14.57	14.57	0.019953	12.49	12.43	29.14
f25	7.28	7.28	0.056234	6.71	6.73	29.14
<i>OS, $H_f = 50$</i>						
100	30.00	28.21	0.000001	28.92	28.21	28.21
75	22.50	22.50	0.007079	18.55	18.33	28.21
50	15.00	15.00	0.017783	13.06	13.01	28.21
f90	25.39	25.39	0.004467	20.96	20.64	28.21
f50	14.11	14.11	0.019953	12.36	12.32	28.21
f25	7.05	7.05	0.063096	6.19	6.22	28.21

Full-variance ESS evaluates the expectation-of-ratios definition with the variance across all historical posterior draws and tree-weight estimate, using 100,000 chi-square quantiles. Block-based ESS is the block average used in the existing scale selector, with 4,000 random scale draws. Differences also reflect numerical integration error; the reciprocal-variance ordering applies when the same integration draws are used. The same calibration is used at $w = 1$ and $w = 0.9$; it remains an all-spike reference.

Table S11(b): MCMC diagnostics for the application log RMST ratio.

Setting	$\hat{R}$	Bulk MCMC ESS	Tail MCMC ESS
<i>PFS</i>			
default, 100	1.0004	7121	7347
default, 75	1.0002	7263	7405
default, 50	1.0003	7478	8092
default, f90	1.0004	7070	7279
default, f50	1.0002	7380	7791
default, f25	1.0002	7677	8093
Hf50, 100	1.0000	6977	7164
Hf50, 75	1.0000	7068	7202
Hf50, 50	1.0001	7234	7730
Hf50, f90	1.0000	6982	7199
Hf50, f50	1.0000	7207	7652
Hf50, f25	1.0000	7537	7848
w0.9, 100	1.0002	7424	7843
w0.9, 75	1.0003	7408	7686
w0.9, 50	1.0005	7396	7826
w0.9, f90	1.0002	7419	7822
w0.9, f50	1.0004	7404	7703
w0.9, f25	1.0006	7407	7716
Hf50_w0.9, 100	0.9999	7562	7559
Hf50_w0.9, 75	0.9999	7553	7673
Hf50_w0.9, 50	0.9998	7572	7687
Hf50_w0.9, f90	0.9999	7561	7559
Hf50_w0.9, f50	0.9998	7567	7650
Hf50_w0.9, f25	0.9999	7638	7928
<i>OS</i>			
default, 100	1.0000	6657	7221
default, 75	1.0000	6812	7697
default, 50	1.0000	6944	7846
default, f90	1.0000	6763	7638
default, f50	1.0000	6966	7886
default, f25	1.0001	7216	7889
Hf50, 100	1.0001	7200	7689
Hf50, 75	1.0003	7283	7849
Hf50, 50	1.0004	7386	7848
Hf50, f90	1.0002	7251	7808
Hf50, f50	1.0004	7405	7769
Hf50, f25	1.0003	7668	7690
w0.9, 100	1.0002	7045	7650
w0.9, 75	1.0001	7109	7849
w0.9, 50	1.0000	7199	7716
w0.9, f90	1.0001	7080	7809
w0.9, f50	1.0000	7213	7500
w0.9, f25	0.9999	7418	7255
Hf50_w0.9, 100	0.9999	7076	7610
Hf50_w0.9, 75	1.0000	7113	7441
Hf50_w0.9, 50	1.0001	7185	7297
Hf50_w0.9, f90	1.0000	7097	7536
Hf50_w0.9, f50	1.0000	7198	7403
Hf50_w0.9, f25	0.9998	7407	7444

Four chains, each with 2,000 warm-up and 2,000 retained draws. These effective sample sizes measure MCMC efficiency and are unrelated to prior ESS. These scalar diagnostics alone do not establish adequate exploration of all tree structures or parameters.

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